

Higher-Arity VC Profiles of Candidate Rejection in Fixed-Interface Clause Systems

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Abstract

The learning algorithm of Shoudai, Matsumoto, Suzuki, and Uchida tests each non-fact fixed-interface candidate clause by searching a Cartesian product of residual substitutions indexed by its distinct variable labels. We isolate the pointwise Boolean candidate-rejection relation underlying this body-positive/head-negative test and determine its higher-arity Cartesian-box shattering profile.

Our main structural conclusion is that candidate rejection has two different sharp information scales. Once a target FICS Γ with at most m clauses is fixed, only polynomially many bounded candidate syntaxes over its active target predicates remain. Hence, for every fixed $1 \leq r \leq k$,

$$d_r(\Gamma) = O((\log(m+1))^{1/r}).$$

This logarithmic scale is simultaneously sharp inside one target: for every fixed $k \geq 2$ we construct a bounded family Γ_m^* such that, for all $1 \leq r \leq k$,

$$d_r(\Gamma_m^*) = \Theta_{k,r}((\log(m+1))^{1/r})$$

at once, using live admissible candidates, and

$$\log_2 |\mathcal{V}_k(\Gamma_m^*)| = \Theta_k(\log(m+1)).$$

The construction stores all required labelings in finite banks of live classifier predicates inside one master target and selects a labeling only by varying the candidate head predicate.

At the global class level, where the target presentation may also vary, the information scale increases by a factor of m . For every fixed bounded FICS regime $\mathfrak{B}$,

$$d_r^{\mathfrak{B}}(m) = O_{\mathfrak{B},k,r}((m \log(m+1))^{1/r}),$$

and for every fixed $k \geq 2$ one bounded regime $\mathfrak{B}_k$, independent of m and r , simultaneously attains the matching order for all $r \leq k$, already in the live-candidate subclass.

The sharp lower bounds require more than finite-class counting. We combine fixed-length residual compression with a linear-size simulation of partial DFAs by rank-two FICSs that preserves the finite context property. Exact length gates confine the rejection relation to the intended finite substitution boxes, while explicit anchored contexts and ordered-boundary positive examples expose the required predicate and residual representations. At any update stage occurring after the corresponding finite exposure threshold, Algorithm 3 literally enumerates the translated candidate and tests the prescribed box. This learner-side realization is conditional on the occurrence of such an update stage.

Thus the candidate-rejection geometry separates a global $m \log m$ information regime, in which the target presentation also varies, from an exact fixed-target $\log m$ regime measuring variation in the target-induced candidate class of one target.

1 Introduction

Learning graph languages requires controlling both the internal structure of individual graph fragments and the ways in which several fragments interact inside grammatical rules. Fixed-interface clause systems (FICSs), introduced by Shoudai, Matsumoto, Suzuki, and Uchida,

provide a logic-programming framework in which such interactions are represented by graph-pattern clauses. For bounded systems satisfying their finite context property (FCP), the learner uses positive presentations and membership queries and admits bounded candidate clauses by testing families of substitutions [14].

The product structure studied here is already explicit in that candidate test. For a non-fact candidate γ , let $X(\gamma) = \{x_1, \dots, x_k\}$ be the set of distinct variable labels. The learner searches substitution families indexed by this set, i.e. a product of the form $R^{X(\gamma)}$, where R is the current residual-representation set. Thus the relevant arity is the number of distinct independently substituted labels, not the number of variable-hyperedge occurrences and not the interface rank of one variable. The Cartesian domain in this paper is therefore not imposed externally on the generated graph language: it is the semantic counterpart of the product search already present in candidate testing.

We evaluate a target-relative candidate $\gamma : q(H) \leftarrow q_1(H_1), \dots, q_\ell(H_\ell)$ against a target FICS Γ . Its *candidate-rejection relation* $V_{\Gamma, \gamma} \subseteq \mathcal{G}_2^k$ labels a compatible substitution tuple positive exactly when every instantiated body atom is refutable in Γ while the instantiated head is not. Shoudai et al. use the existence of a body-positive/head-negative substitution to exclude a spurious candidate. We study the full pointwise Boolean relation behind that existential test. In this sense the paper asks for the geometry of rejection witnesses rather than only whether at least one witness exists.

The word “semantic” is used below only in the limited, target-relative sense that labels are determined by refutability in the target clause system rather than by candidate syntax alone. The relation is nevertheless *presentation-sensitive*: equivalent FICS presentations of the same start language may have different internal predicates and different candidate interactions. Accordingly, we use *candidate-rejection relation* as the primary term and do not treat it as an invariant of the generated language $L(\Gamma, p)$. Actual target clauses themselves have empty rejection relations.

FCP supplies the operational bridge to the learner. Throughout this paper, *FCP* refers exclusively to the predicate-level finite context property for FICSs in [14]. The terminology belongs to the broader distributional-learning tradition, but we do not identify this condition with the strong, weak, or very weak finite-context properties studied for string grammars, nor do we assert implications between those notions [8]. On a context-compatible domain, the FICS-FCP reduces each predicate-level test to membership in the distinguished start language. Once the corresponding context representatives occur in the learner’s predicate basis, replacing target predicates by their representative predicates turns a live target-relative candidate into an actual bounded candidate of the learner, and its body-positive/head-negative test agrees pointwise with $V_{\Gamma, \gamma}$.

The FGS line preceding the present FICS framework also has a direct connection to computational learning theory. Shoudai, Matsumoto, Suzuki, Uchida, and Miyahara [13] proved polynomial-time PAC learnability for a parameterized class of variable-hereditary FGS languages under bounded-treewidth and bounded-degree restrictions. Their PAC problem is an ordinary graph-language problem: examples are whole graphs and the learned concept is a graph language. The capacity studied here is different. We do not measure the ordinary VC dimension or sample complexity of the FICS language class itself; instead, for a fixed candidate arity k , we study the higher-arity Cartesian-box shattering behavior of the internal candidate-rejection relations on tuples of residual substitutions. Thus the two viewpoints are complementary: the earlier result concerns PAC learnability of graph-language classes, whereas the present results quantify the combinatorial interaction capacity of candidate tests occurring inside a distributional FICS learner. No implication between the two kinds of bounds is asserted here.

A related line of distributional-learning work studies grammar classes relative to a fixed

finite observation. For context-free grammars, Kuriyama [11] considers substitutability under a fixed explicit finite-monoid typing and develops finite typed reconstruction from positive data. More recently, the same fixed-observation viewpoint has been extended to bounded-fan-out linear MCFG presentations, where an explicit finite monoid homomorphism is supplied uniformly across each observation fiber [12]. These results concern reconstruction and learnability rather than shattering capacity, but they motivate a broader question pursued here: once a grammar-learning procedure reduces candidate testing to finitely represented observations, what combinatorial complexity remains in the interaction among several simultaneously substituted components? The fixed-observation restriction in these earlier learning results should not be confused with the fixed-target regime studied below: the former fixes an auxiliary finite observation uniformly across a language class, whereas the latter conditions candidate-rejection capacity on one target presentation.

The resulting relations are genuinely k -ary. The conceptual starting point of the present paper is the product-space viewpoint developed in the higher-dimensional PAC_n -learning framework of Kobayashi, Kuriyama, and Takeuchi [9]. We apply that viewpoint here to a concrete grammatical-inference mechanism. In the FICS learner this Cartesian structure is not imposed only as an external formal analogy: distinct variable labels in one candidate clause give rise to independently chosen residual substitutions, so the learner itself searches a genuine product domain. Candidate rejection therefore provides a natural place to ask for the corresponding higher-arity shattering geometry.

Takeuchi later investigated the associated VC_2 -dimension viewpoint [15]. Higher-arity Cartesian-box VC notions, closely related to the parameter used below, also occur in the hypergraph and model-theoretic literature; we use Chernikov and Towsner [1] as a principal reference for this higher-arity VC viewpoint. A distinct recent line develops high-arity learning for graphs, hypergraphs, and relational structures [4–6]. There is also recent discussion concerning the relation between higher-arity VC notions and PAC_n -type learnability in product spaces [2, 10]. The present paper uses this product-space perspective as motivation and a combinatorial language, but its results are capacity theorems for FICS candidate-rejection relations, not a general PAC_n characterization. In particular, none of the proofs below depends on such a characterization, and this paper does not settle the general PAC_n characterization problem posed in [15].

Our contribution is not a new abstract shattering dimension. The novelty is also not the appearance of an $m \log m$ counting scale in isolation. For bounded presentations, such a scale is compatible with the number of available canonical descriptions, and related $n \log n$ -type capacity phenomena are familiar from finite-state concept classes. The problem here is to determine whether that information scale can actually be realized by the highly constrained candidate tests of bounded FICSs while preserving fixed interfaces, predicate liveness, FCP, and the representative connection used by the learner. We prove that it can, and we further show that fixing the target collapses the sharp information scale from $m \log m$ to $\log m$. Thus we determine the Cartesian-box shattering profile of the concrete candidate-rejection relations induced by bounded FICS presentations rather than introduce a new abstract dimension.

There is also a distinct tradition of tuple-based distributional and substitutability learning for multiple context-free formalisms. Yoshinaka’s multidimensional substitutability, Clark–Yoshinaka’s distributional learning of parallel MCFGs, and recent fixed-observation learning of bounded-fan-out linear MCFG presentations [3, 12, 16] treat tuples generated by multi-dimensional nonterminals. The arity here is of a different kind: $k = |X(\gamma)|$ counts several independently substituted graph variables occurring simultaneously in one candidate clause. Thus it measures interaction among substitution slots rather than fan-out of one derived tuple.

To state the capacities, we use the bounded FICS notion of Shoudai et al. [14] without

modification. We fix Δ, s, t, w, d , the finite label alphabets, and the finite admissible head-frame family $\mathcal{H}_d$, and allow only the clause budget m to vary. We denote this fixed presentation-level subregime by $\mathfrak{B}$. Thus the target systems at budget m form a fixed subregime of the FCP class $\mathcal{FICSL}_{\Delta}^{\text{FCP}}(m, s, t, w, d)$ of [14]. Let $\mathcal{V}_k^{\mathfrak{B}}(m)$ be the class of candidate-rejection relations induced by such presentations and by admissible candidates with exactly k distinct rank-two variable labels. For $1 \leq r \leq k$, let $d_r^{\mathfrak{B}}(m)$ be the largest r -box side length obtainable after fixing $k - r$ coordinates.

Scope of the candidate classes. It is important to distinguish the target-induced candidate classes studied here from the full sample-dependent candidate space generated by the learner. For a fixed target presentation Γ , our class $\mathcal{V}_k(\Gamma)$ varies only admissible candidates whose predicate symbols belong to the active target predicate universe $\text{Pred}^*(\Gamma)$. At a concrete learner stage, by contrast, Algorithm 3 forms bounded candidates over the current predicate basis $F \cup \{\rho_{\emptyset}\}$, and F may contain boundary representatives that do not correspond to predicates of the target presentation.

Accordingly, the fixed-target results below do not bound the capacity of the full sample-induced candidate space over F . They measure the target-induced subfamily determined by one target presentation. For live target predicates, representative translation embeds this subfamily into the actual learner candidate space at stages where the required representatives have been exposed. The learner-side realization results later in the paper make this connection explicit.

Main Theorem A: target-induced fixed-target capacity. We first fix the target presentation and vary only admissible candidates over its active predicate universe. This isolates the information carried by candidate choice within the target-induced candidate class, as opposed to the additional information that can be encoded by varying the target presentation. For every target Γ of clause budget at most m ,

$$d_r(\Gamma) = O_{\mathfrak{B},k,r}((\log(m+1))^{1/r}).$$

For every fixed $k \geq 2$, this is simultaneously sharp for all $1 \leq r \leq k$ inside one bounded target family:

$$d_r(\Gamma_m^*) = \Theta_{k,r}((\log(m+1))^{1/r}).$$

Moreover,

$$\log_2 |\mathcal{V}_k(\Gamma_m^*)| = \Theta_k(\log(m+1)).$$

Thus one master target can contain enough live classifier predicates to realize the entire sharp fixed-target profile at once.

Main Theorem B: global class-level capacity. If the target presentation is also allowed to vary, then for every fixed bounded regime

$$\log_2 |\mathcal{V}_k^{\mathfrak{B}}(m)| = O_{\mathfrak{B},k}(m \log(m+1))$$

and

$$d_r^{\mathfrak{B}}(m) = O_{\mathfrak{B},k,r}((m \log(m+1))^{1/r}).$$

For every fixed $k \geq 2$, one bounded rich regime $\mathfrak{B}_k$, independent of m and r , simultaneously realizes

$$d_r^{\mathfrak{B}_k}(m) = \Theta_{k,r}((m \log(m+1))^{1/r})$$

for all $1 \leq r \leq k$, already in the live-candidate subclass, and

$$\log_2 |\mathcal{V}_k^{\mathfrak{B}_k}(m)| = \Theta_k(m \log(m+1)).$$

The word “simultaneously” here refers to the common ambient regime: the extremal target–candidate pair may depend on r , and the target also varies across the labelings required by shattering.

Equivalently, the two sharp information levels are

$$(d_r(\Gamma_m^*))^r \asymp \log m \quad \text{and} \quad (d_r^{\mathfrak{B}_k}(m))^r \asymp m \log m.$$

The first measures candidate variation conditional on one target; the second also permits the target presentation itself to encode information. Both levels are target-induced: admissible candidate predicates are tied to the active predicates of the corresponding target presentation. They should therefore be distinguished from the potentially larger, sample-dependent candidate space generated by Algorithm 3 over the current learner basis F .

Contributions. First, we isolate the k -ary candidate-rejection relation implicit in the Cartesian residual search of Algorithm 3 and define its fiber-wise box profile. Second, we prove the two sharp capacity theorems above. Third, we show that both scales are attainable by live candidates inside bounded FCP-FICSs. The lower-bound transfer combines fixed-length residual compression with a linear partial-DFA-to-rank-two-FICS simulation preserving FCP, exact length gating, anchored contexts, and ordered-boundary residual exposure. Fourth, we construct one master target containing classifier banks that simultaneously attains every target-induced fixed-target fiber profile $1 \leq r \leq k$, separating information encoded by the target presentation from information encoded by the choice of a target-relative candidate.

The proofs separate counting from realization. Counting yields the two entropy ceilings: $O(m \log m)$ globally and $O(\log m)$ once the target is fixed. The nontrivial part is to attain these ceilings within the exact FICS candidate geometry. Fixed-length residual compression encodes arbitrary box labelings into partial DFAs of size $O(D^r / \log D)$, and an FCP-preserving linear-size simulation transfers those finite-state descriptions into rank-two FICS syntax. Length gates restrict rejection exactly to the intended finite substitution boxes, while anchored contexts and ordered-boundary positive examples expose the predicates and residuals needed for learner-side realization. For the fixed-target theorem, a single master automaton additionally contains banks of live classifier roots for all $1 \leq r \leq k$, so all sharp fiber profiles coexist inside one target.

The role of finite automata here is not to provide a new finite-state VC theorem. Classical automata-capacity results already exhibit closely related $n \log n$ -type behavior. What we require is instead an explicit fixed-length encoding of arbitrary box labelings that can be transferred, with linear overhead and FCP preservation, into the exact rank-two FICS syntax used below.

Organization. Section 2 defines bounded regimes, live predicates, candidate-rejection relations, the boundary-representation machinery used by the learner, FCP observation reduction, the direct Algorithm 3 bridge, and the fiber-wise box-VC profile. Section 3 proves global and fixed-target counting bounds. Section 4 develops residual DFA compression and an FCP-preserving DFA-to-FICS transfer, constructs the sharp global lower bound, and then builds a single master target with simultaneously sharp fixed-target profiles and an explicit learner-side realization. Section 5 focuses on interpretation of the two information scales and the open structural and algorithmic questions.

2 Preliminaries

We recall only the parts of fixed-interface clause systems and higher-arity shattering needed later. We follow the FICS framework of [14]. Numbered cross-references to that preprint refer to arXiv:2604.26333v2. Throughout, all label alphabets are finite.

2.1 Graphs with interfaces and substitution

A *graph with interface* is a finite labeled graph together with an ordered tuple $\iota_G = (v_1, \dots, v_r)$ of pairwise distinct vertices. The integer r is its interface rank. Isomorphisms preserve labels and the order of interface vertices.

A graph pattern may contain variable hyperedges. Each variable hyperedge has an ordered list of ports. A rank- r variable may be replaced only by a rank- r graph with interface. If $\theta = \{x_1 := K_1, \dots, x_s := K_s\}$ is compatible with a pattern H , then $\text{Real}(H, \theta)$ denotes the realization obtained by replacing each occurrence of x_i with a copy of K_i and identifying ordered interface vertices with ordered ports. If one variable label occurs repeatedly, all of its occurrences use copies of the same substituted graph.

For compatible rank- r objects we write $C \odot K$ for the interface composition of [14]. Concretely, one first takes disjoint copies and then identifies corresponding interface vertices. Non-interface vertices are never identified across the two components. Ordinary edges can acquire the same pair of endpoints only as a consequence of these interface identifications, in which case the compatibility condition in the composition definition applies.

2.2 Fixed-interface clauses

A fixed-interface clause has the form $q(H) \leftarrow q_1(H_1), \dots, q_\ell(H_\ell)$, subject to the fixed-interface correspondence between head-variable occurrences and body star patterns in [14]. Write $\text{Occ}_H(\gamma)$ for the set of variable occurrences in the head of a clause or candidate γ , and let $\lambda : \text{Occ}_H(\gamma) \rightarrow X(\gamma)$ map each occurrence to its variable label. The fixed-interface correspondence is occurrence-based, whereas the substitution family below is indexed by the distinct labels in $X(\gamma)$. If several occurrences carry the same label, they receive copies of the same substituted graph.

A finite set of fixed-interface clauses is a FICS. We distinguish a start predicate p and write $\Gamma \models q(K)$ when the goal $q(K)$ is refutable in Γ . The generated language is $L(\Gamma, p) = \{G : \Gamma \models p(G)\}$.

2.3 Bounded systems and fixed regimes

We use the boundedness notion of Shoudai et al. [14] as stated in their Definition 2.25, including the finite head-frame condition, and spell out all components of the bounded regime that are used in the present capacity arguments. For nonnegative integers m, s, t, w, d , an (m, s, t, w, d) -*bounded FICS* is a fixed-interface clause system satisfying Definition 2.25 of [14]: it has at most m clauses; every variable label has rank at most w ; and, in every clause, the head has at most s variable-hyperedge occurrences, the body has at most t atoms, every graph pattern has maximum vertex degree at most d , and the head frame belongs to the fixed finite family $\mathcal{H}_d$ of admissible head-frame types. The vertex- and edge-label alphabets Σ_V, Σ_E are finite and fixed, as in the source framework. In particular, the finite head-frame condition is part of the source boundedness notion itself (Definition 2.25(d) and Remark 2.26 of [14]); it is not an additional restriction introduced here.

For a semantic degree bound Δ , Shoudai et al. write $\mathcal{FICS}_{\Delta}(m, s, t, w, d)$ for the

corresponding generated graph-language class, and $\mathcal{FICS}\mathcal{L}_\Delta^{\text{FCP}}(m, s, t, w, d)$ for its finite-context-property subclass.

In this paper the clause budget m is the only asymptotic parameter. Accordingly, we use the following terminology.

Definition 2.1 (Fixed bounded regime). A *bounded FICS regime* $\mathfrak{B}$ is a fixed choice of $\Delta, s, t, w, d, \Sigma_V, \Sigma_E$, and the finite admissible head-frame family $\mathcal{H}_d$, all independent of m . At clause budget m , the targets admitted by $\mathfrak{B}$ are precisely FCP-FICS presentations satisfying these fixed parameters and having at most m clauses. Thus $\mathfrak{B}$ introduces no new boundedness condition; it only freezes all parameters of the bounded class of [14] except the clause budget m .

Because candidate rejection is presentation-sensitive, $\mathfrak{B}$ is used here at the presentation level even though $\mathcal{FICS}\mathcal{L}_\Delta^{\text{FCP}}(m, s, t, w, d)$ is notation for a class of generated graph languages.

All constants in regime-uniform upper bounds may depend on the fixed data of $\mathfrak{B}$ and on the fixed arity parameters, but not on m . Constants in the explicit lower-bound regimes constructed later may depend on k, r , but not on m, D , or the target labeling S .

For a FICS Γ with distinguished start predicate p , define its active predicate universe by $\text{Pred}^*(\Gamma) = \{p\} \cup \{q : q \text{ occurs as a head or body predicate in a clause of } \Gamma\}$. An admissible candidate γ over Γ is required to satisfy the same local bounds and fixed head-frame restriction as clauses of the ambient regime, to satisfy $\text{Pred}(\gamma) \subseteq \text{Pred}^*(\Gamma)$, and, in the concept class studied here, to have exactly k distinct variable labels, all of interface rank two. Variable names are considered only up to rank-preserving alpha-renaming.

Define also the *live predicate universe* $\text{Pred}^+(\Gamma) = \{q \in \text{Pred}^*(\Gamma) : \exists K \Gamma \models q(K)\}$. An admissible candidate is *live* if every predicate occurring in it belongs to $\text{Pred}^+(\Gamma)$. The live restriction is not needed for the global counting upper bound; its role is to isolate the target-relative candidates for which the representative connection used in the learning proof can eventually be exposed by positive data.

For a fixed regime $\mathfrak{B}$, write $\mathcal{V}_k^{\mathfrak{B}}(m)$ for the candidate-rejection class defined below using FCP-FICSs in $\mathfrak{B}$ with at most m clauses, and write $\mathcal{V}_{k,\text{live}}^{\mathfrak{B}}(m)$ for the subclass obtained by requiring the candidate to be live.

2.4 Finite context property

Throughout this paper, *FCP* means the finite context property of the fixed-interface clause-system framework of [14]. We do not identify this predicate-level graph condition with the strong, weak, or very weak finite-context properties studied for context-free grammar learning [8]; no implication between those notions is used here.

A rank- r predicate q has the FICS finite context property if there exists a rank- r context C_q such that, for every rank- r ground graph K for which the composition $C_q \odot K$ is defined,

$$\Gamma \models q(K) \iff \Gamma \models p(C_q \odot K).$$

A system satisfies FCP if all predicates admit such contexts. FCP is used below as an observation reduction and as the bridge to learner representatives; it is not part of the definition of the rejection relation itself.

2.5 Candidate-rejection relations

For each $r \geq 0$, let $\mathcal{G}_r = \{\text{rank-}r \text{ ground graph patterns with ordered interface}\} / \cong_\ell$ denote interface-preserving isomorphism classes. We use the same symbol for a class and a chosen representative when no confusion can arise.

Let $\gamma : q(H) \leftarrow q_1(H_1), \dots, q_\ell(H_\ell)$ be an admissible candidate whose distinct variable labels are $X(\gamma) = \{x_1, \dots, x_k\}$. By admissibility, every x_i has interface rank two, so the common domain is $\mathcal{G}_2^k$. For $\bar{K} = (K_1, \dots, K_k) \in \mathcal{G}_2^k$ write $\theta_{\bar{K}} = \{x_1 := K_1, \dots, x_k := K_k\}$. Every occurrence $o \in \text{Occ}_H(\gamma)$ carrying label $\lambda(o) = x_i$ receives a copy of the same graph K_i .

Definition 2.2 (Candidate-rejection relation). The target-relative relation $V_{\Gamma, \gamma} \subseteq \mathcal{G}_2^k$ is defined by $V_{\Gamma, \gamma}(\bar{K}) = 1$ iff:

- (i) $\text{Real}(H, \theta_{\bar{K}})$ is defined;
- (ii) every instantiated body atom is refutable in Γ ; and
- (iii) the realized head is not refutable in Γ .

Thus $V_{\Gamma, \gamma}$ records body-positive/head-negative substitutions. Its labels are semantic in the target-relative sense of refutability, but the relation is presentation-sensitive and is not determined by the generated start language alone.

Lemma 2.3 (Target clauses have empty rejection relations). *If $\gamma \in \Gamma$, then $V_{\Gamma, \gamma} = \emptyset$.*

Proof. For every compatible ground substitution, refutability of all instantiated body atoms together with the clause $\gamma \in \Gamma$ yields refutability of the instantiated head. Hence no body-positive/head-negative substitution exists. $\square$

For a fixed bounded regime $\mathfrak{B}$, define

$$\mathcal{V}_k^{\mathfrak{B}}(m) = \left\{ V_{\Gamma, \gamma} : \begin{array}{l} (\Gamma, p) \text{ satisfies FCP and the fixed bounded parameters of } \mathfrak{B}, \\ \Gamma \text{ has at most } m \text{ clauses,} \\ \gamma \text{ is an admissible bounded candidate over } \Gamma, |X(\gamma)| = k, \\ \text{rank}(x) = 2 \text{ for every } x \in X(\gamma) \end{array} \right\}.$$

Equivalently, the targets are presentations whose generated languages lie in the fixed subregime of $\mathcal{FICSL}_{\Delta}^{\text{FCP}}(m, s, t, w, d)$ specified by $\mathfrak{B}$, while $\mathcal{V}_k^{\mathfrak{B}}(m)$ retains the target presentation because candidate rejection is presentation-sensitive.

The live subclass $\mathcal{V}_{k, \text{live}}^{\mathfrak{B}}(m)$ is defined by adding $\text{Pred}(\gamma) \subseteq \text{Pred}^+(\Gamma)$. Plainly, $\mathcal{V}_{k, \text{live}}^{\mathfrak{B}}(m) \subseteq \mathcal{V}_k^{\mathfrak{B}}(m)$.

2.6 Boundary representations and learner stages

We briefly recall the parts of the learner of Shoudai et al. [14] that are needed to connect the candidate-rejection relations above with actual candidate tests. The purpose is only to fix the notation and the information flow used later; the complete learning algorithm and its correctness proof remain those of [14].

Let

$$D_n = \{G_1, \dots, G_n\}$$

be the finite positive sample available at stage n . For a positive graph G , an ordered boundary specification consists of an ordered tuple $\beta = (b_1, \dots, b_r)$ of distinct boundary vertices together with a set E_B of graph edges incident with those boundary vertices. When the specification is valid in the sense of Definition 2.5 of [14], it determines a unique subgraph with ordered interface

$$K_G(\beta, E_B).$$

A boundary representation ρ records such finite boundary data arising from a positive example, and we write

$$\text{can}(\rho)$$

for its canonically determined graph with interface. The rank of ρ is the size of its ordered interface.

The learner enumerates all valid boundary representations of rank at most w induced by the current positive sample. We denote this finite set by

$$\text{BRep}_w(D_n).$$

Thus $\text{BRep}_w(D_n)$ is a sample-induced collection of graph fragments with ordered interfaces, represented syntactically by boundary data and interpreted through can .

Two sets of boundary representations play different roles. The *residual set* is refreshed from the current positive sample and, in the regime used below, is

$$R = \text{BRep}_w(D_n).$$

The *predicate basis* F , by contrast, changes only when the outer learning procedure performs a basis update. We call such a stage an *update stage*; at such a stage the learner replaces its predicate basis by the current boundary-representation set,

$$F = \text{BRep}_w(D_n).$$

Between update stages the previous predicate basis is retained even though the residual set is refreshed from the growing positive sample.

For every representation $\rho \in F \cup \{\rho_\emptyset\}$, where $\rho_\emptyset$ is the distinguished rank-zero representation, Algorithm 3 introduces a learner predicate symbol p_ρ of the same interface rank. The representation therefore has a syntactic role, as an index of a learner predicate, and a semantic role through the graph with interface $\text{can}(\rho)$. Membership queries for defined compositions

$$\text{can}(\rho) \odot \text{can}(\sigma), \quad \rho \in F \cup \{\rho_\emptyset\}, \sigma \in R,$$

provide the observation information used by Algorithm 3 to evaluate candidate body and head atoms.

Only one part of Algorithm 3 is needed in the present paper. Over the learner predicates indexed by $F \cup \{\rho_\emptyset\}$, the algorithm enumerates all bounded fixed-interface clause candidates with admissible head frames. For a non-fact candidate δ , let $X(\delta)$ be its set of distinct variable labels. Algorithm 3 then ranges over the Cartesian product

$$(\sigma_x)_{x \in X(\delta)} \in R^{X(\delta)}.$$

For a tested family it uses the canonical residual graphs

$$K_x = \text{can}(\sigma_x) \quad (x \in X(\delta)),$$

with the same residual assigned to every occurrence carrying the same variable label. If all instantiated body observations are positive, the algorithm performs the corresponding head membership test; a negative head answer marks the candidate inadmissible. Hence the Cartesian product $R^{X(\delta)}$ is not an auxiliary construction introduced here: it is the product search whose pointwise body-positive/head-negative geometry is studied by the candidate-rejection relation.

Finally, target predicates and learner predicates are different syntactic objects. If a target predicate q has an FCP context C_q , a boundary representation $\rho_q \in F$ satisfying

$$\text{can}(\rho_q) = C_q$$

is called a *representative* of q at that learner stage. For a live target predicate, the representative-exposure results of [14] imply that, after a finite positive-sample threshold, an appropriate boundary representation occurs in $\text{BRep}_w(D_n)$, and at the sufficiently late update stages covered by the corresponding context-exposure argument it belongs to F . This is the mechanism used below to translate target-relative candidates into actual learner candidates.

2.7 Observation reduction and Algorithm-3 realization

The rejection relation is defined using target predicates, whereas FCP provides an observation reduction only when the required context compositions are defined. For each predicate q' occurring in γ , fix one witnessing context $C_{q'}$. For a substitution tuple $\bar{K}$, let $H[\bar{K}]$ and $H_i[\bar{K}]$ denote the realized head and body arguments whenever defined.

Define the *context-compatible domain*

$$\Omega_{\Gamma, \gamma, C} = \left\{ \bar{K} \in \mathcal{G}_2^k : \begin{array}{l} H[\bar{K}] \text{ and all } H_i[\bar{K}] \text{ are defined,} \\ C_q \odot H[\bar{K}] \text{ is defined,} \\ C_{q_i} \odot H_i[\bar{K}] \text{ is defined for every } i \end{array} \right\}.$$

On this domain define the observable rejection relation $V_{\Gamma, \gamma, C}^{\text{obs}} : \Omega_{\Gamma, \gamma, C} \rightarrow \{0, 1\}$ by $V_{\Gamma, \gamma, C}^{\text{obs}}(\bar{K}) = 1$ iff $\Gamma \models p(C_{q_i} \odot H_i[\bar{K}])$ ($1 \leq i \leq \ell$), and $\Gamma \not\models p(C_q \odot H[\bar{K}])$.

Lemma 2.4 (FCP observation reduction). *For every choice of witnessing FCP contexts,*

$$\boxed{V_{\Gamma, \gamma} \upharpoonright \Omega_{\Gamma, \gamma, C} = V_{\Gamma, \gamma, C}^{\text{obs}}.}$$

Thus, on context-compatible substitution tuples, every predicate-level test occurring in γ is reducible to membership in the start language.

Proof. For every body predicate q_i , FCP gives $\Gamma \models q_i(H_i[\bar{K}]) \iff \Gamma \models p(C_{q_i} \odot H_i[\bar{K}])$ for every $\bar{K} \in \Omega_{\Gamma, \gamma, C}$, and the same equivalence holds for the head predicate. Substitution into the definition of $V_{\Gamma, \gamma}$ proves the claim. $\square$

Remark 2.5 (Role of the context-compatible domain). The domain $\Omega_{\Gamma, \gamma, C}$ is not part of the definition of the candidate-rejection concept. It is used only when target-level refutability is translated into start-language observations through FCP. Consequently, the abstract counting and VC upper bounds below require no context choice. In the lower-bound construction, however, we verify separately that the entire extremal box lies in the relevant context-compatible domain before invoking the learner-side interpretation.

The target-predicate candidate and the learner candidate live over different predicate alphabets. Let F and R be the predicate basis and residual set described in Section 2.6. Suppose that for every predicate q' occurring in γ , the learner has a representative $\rho_{q'} \in F \cup \{\rho_\emptyset\}$, $\text{can}(\rho_{q'}) = C_{q'}$. For the distinguished rank-zero start predicate we may take the special representation $\rho_\emptyset$. Define the *representative translation*

$$\tau_\rho(\gamma) : p_{\rho_q}(H) \leftarrow p_{\rho_{q_1}}(H_1), \dots, p_{\rho_{q_\ell}}(H_\ell).$$

Only predicate symbols are changed; head frames, body stars, variable labels, occurrences, ranks, and the occurrence-based fixed-interface correspondence are unchanged.

Theorem 2.6 (Direct realization in Algorithm 3). *Let (Γ, p) be an FCP-FICS and let γ be a bounded live admissible candidate. Consider a stage of the learner of [14] with current predicate basis F and residual set R , and suppose that representatives $\rho_{q'} \in F \cup \{\rho_\emptyset\}$ as above are available for every predicate occurring in γ .*

Then the representative translation $\tau_\rho(\gamma)$ belongs to the bounded candidate set $\mathcal{C}$ constructed by Algorithm 3 of [14].

Moreover, let $(\sigma_x)_{x \in X(\gamma)} \in R^{X(\gamma)}$ be a residual family examined by the Cartesian-product loop of Algorithm 3, put $K_x = \text{can}(\sigma_x)$, and suppose that its canonical tuple lies in $\Omega_{\Gamma, \gamma, C}$. Then the non-fact clause-admission test of Algorithm 3 rejects this family if and only if

$$V_{\Gamma, \gamma}((K_x)_{x \in X(\gamma)}) = 1.$$

Proof. By Section 2.6, Algorithm 3 enumerates bounded fixed-interface candidates over $F \cup \{\rho_\emptyset\}$ and tests them over the product of current residual representations indexed by their distinct variable labels. More precisely, because γ is bounded and admissible, its head frame belongs to the fixed family $\mathcal{H}_d$, and its body length, variable ranks, variable-hyperedge occurrences, and local graph patterns satisfy the fixed bounds of the ambient regime. By assumption, $\rho_q, \rho_{q_1}, \dots, \rho_{q_\ell} \in F \cup \{\rho_\emptyset\}$. Algorithm 3 of [14] first introduces the corresponding predicate symbols and then constructs $\mathcal{C}$ as the set of all bounded fixed-interface clause candidates over $F \cup \{\rho_\emptyset\}$ whose head frames belong to $\mathcal{H}_d$. Representative translation changes only predicate symbols and preserves every other syntactic datum of γ . Consequently, $\tau_\rho(\gamma) \in \mathcal{C}$.

Now fix a family $(\sigma_x)_{x \in X(\gamma)} \in R^{X(\gamma)}$. The Cartesian-product loop of Algorithm 3 explicitly ranges over all such families and proceeds to the head test exactly when all body observations are positive. On the context-compatible domain, the observation table and FCP identify these positive body observations with refutability of the corresponding instantiated target body atoms, by Lemma 2.4.

Algorithm 3 then sets $K_x = \text{can}(\sigma_x)$ ($x \in X(\gamma)$), and realizes the head with this same label-indexed substitution. Thus repeated occurrences carrying the same variable label receive the same canonical residual graph, exactly as in the definition of $V_{\Gamma, \gamma}$. The algorithm then asks the head membership query with the context $\text{can}(\rho_q) = C_q$. Again by FCP and Lemma 2.4, a negative answer is equivalent, on $\Omega_{\Gamma, \gamma, C}$, to non-refutability of the instantiated target head. Hence the non-fact admission test marks the candidate inadmissible precisely for those families for which all target body atoms are refutable and the realized target head is not refutable. This is exactly the condition $V_{\Gamma, \gamma}((K_x)_x) = 1$. $\square$

Remark 2.7 (Availability of representatives for live predicates). The representative hypothesis in Theorem 2.6 is a literal condition on the learner state. As summarized in Section 2.6, if a target predicate q' is live, then Lemma 4.15 of [14] supplies, after a finite positive- sample threshold, an ordered boundary representation

$$\rho_{q'} \in \text{BRep}_w(D_n)$$

whose canonical graph is an FCP context $C_{q'}$. At a sufficiently late update stage satisfying the context condition, Lemma 4.17 places this representative in the current predicate basis F . The special $\rho_\emptyset$ handles the distinguished rank-zero start context. Thus liveness is the nonempty-extension condition under which the representative- exposure mechanism applies.

Remark 2.8. The theorem does not assert a literal inclusion between the learner's candidate set and the target-predicate candidate set: the two use different predicate alphabets. It gives a direct translation at stages where the required non-start representatives belong to F , with the special $\rho_\emptyset$ available for the distinguished rank-zero start context. Conversely, the learner may contain boundary representatives that do not correspond to target predicates, so a converse translation is not claimed.

Remark 2.9. The relevant arity is $|X(\gamma)|$, the number of distinct independently substituted variable labels. This agrees with the product index used by the candidate test. It is distinct from the number of variable occurrences and from MCFG fan-out or tuple dimension.

2.8 Box shattering and fibers

Let $\mathcal{C} \subseteq 2^{X_1 \times \dots \times X_r}$. A product $A_1 \times \dots \times A_r$, $|A_i| = D$, is a D -box. It is shattered if every subset of its D^r cells is realized by the restriction of a concept in $\mathcal{C}$. Let $\text{VC}_r^\square(\mathcal{C})$ be the largest such D , with value $+\infty$ if arbitrarily large boxes are shattered. For $r = 1$, this is the ordinary VC dimension.

Remark 2.10 (Relation to higher-arity VC notation). The parameter $\text{VC}_r^\square$ is the side-length form of the standard r -dimensional Cartesian-box shattering parameter for r -ary relations used in higher-arity VC theory; see, for example, Chernikov and Towsner [1]. We retain the superscript $\square$ only to emphasize that the shattered set is a Cartesian box rather than an arbitrary subset of the product domain.

For $I \subseteq [k]$ and a tuple a fixing the coordinates in I , let $C_{I=a}$ denote the corresponding fiber. For a bounded regime $\mathfrak{B}$, define

$$d_r^\mathfrak{B}(m) = \sup_{\substack{I \subseteq [k] \\ |I|=k-r}} \sup_a \text{VC}_r^\square((\mathcal{V}_k^\mathfrak{B}(m))_{I=a}), \quad 1 \leq r \leq k.$$

In particular, $d_k^\mathfrak{B}(m) = \text{VC}_k^\square(\mathcal{V}_k^\mathfrak{B}(m))$. For $r = 1$, $d_1^\mathfrak{B}(m)$ is exactly the maximum ordinary VC dimension of a one-coordinate fiber obtained after fixing the remaining $k - 1$ coordinates. Thus it is the natural fiber/graph-VC parameter of the underlying k -ary relation class.

Define the live profile analogously by

$$d_{r,\text{live}}^\mathfrak{B}(m) = \sup_{\substack{I \subseteq [k] \\ |I|=k-r}} \sup_a \text{VC}_r^\square((\mathcal{V}_{k,\text{live}}^\mathfrak{B}(m))_{I=a}).$$

Since the live class is a subclass, $d_{r,\text{live}}^\mathfrak{B}(m) \leq d_r^\mathfrak{B}(m)$.

3 Counting and Upper Bounds

The arguments in this section are combinatorial and do not use FCP.

3.1 Canonical predicate names

Lemma 3.1 (Active predicate bound). *For every FICS Γ with at most m clauses,*

$$|\text{Pred}^*(\Gamma)| \leq m(t + 1) + 1.$$

Proof. Every clause contains one head predicate and at most t body predicates. Hence an m -clause system has at most $m(t + 1)$ predicate occurrences. The active predicate universe contains only predicates occurring in clauses, together with the distinguished start predicate, giving the stated bound. $\square$

Let $R_\mathfrak{B}$ be the finite set of predicate ranks permitted by the fixed regime $\mathfrak{B}$, and put $M_m = m(t + 1) + 1$. For each $\rho \in R_\mathfrak{B}$, let $\Pi_{m,\rho} = \{\pi_{\rho,1}, \dots, \pi_{\rho,M_m}\}$ be fresh predicate symbols of rank ρ , and define the rank-tagged pool

$$\Pi_m = \bigsqcup_{\rho \in R_\mathfrak{B}} \Pi_{m,\rho}.$$

Because $R_\mathfrak{B}$ is fixed, $|\Pi_m| = |R_\mathfrak{B}|M_m = O_\mathfrak{B}(m)$.

Lemma 3.2 (Rank-tagged canonical renaming). *Let Γ be a FICS in $\mathfrak{B}$ with at most m clauses. There is an injective rank-preserving map $\eta : \text{Pred}^*(\Gamma) \rightarrow \Pi_m$. Applying η simultaneously to the target system and to an admissible candidate preserves refutability and hence preserves the induced candidate-rejection relation up to the canonical predicate renaming.*

Proof. For every permitted rank ρ , the number of active predicates of rank ρ is at most M_m by Lemma 3.1; map them injectively into $\Pi_{m,\rho}$. Taking the disjoint union of these maps gives η . A uniform renaming of predicate symbols in both the clause system and the candidate induces a bijection between refutations before and after renaming, so every body and head refutability test, and therefore $V_{\Gamma,\gamma}$, is preserved. $\square$

By Lemma 3.2, every target/candidate pair may be represented over the fixed rank-tagged pool Π_m of size $O_{\mathfrak{B}}(m)$.

3.2 Counting bounded clauses

Lemma 3.3. *There is a constant $c_{\text{cl}} > 0$ such that the number $N_\ell(P)$ of bounded fixed-interface clauses with exactly ℓ body atoms over a predicate pool of size P satisfies $N_\ell(P) \leq c_{\text{cl}} P^{\ell+1}$, $0 \leq \ell \leq t$.*

Proof. The head predicate and the ℓ body predicates contribute at most $P^{\ell+1}$ choices. By Definition 2.1, the remaining syntactic choices are governed by the fixed bounded parameters of [14]. In particular, the head frame must belong to the fixed finite family $\mathcal{H}_d$. For each such frame, the bounds s, t, w, d , together with the fixed finite alphabets Σ_V, Σ_E , leave only constantly many variable-label equality patterns, body-star patterns, rank assignments, and fixed-interface occurrence correspondences. Hence all non-predicate choices are bounded by a constant depending only on the fixed regime $\mathfrak{B}$, and can be absorbed into c_{cl} . $\square$

This is the static counterpart of the candidate-counting argument in the FICS learning algorithm [14].

Let Q_m be the total number of bounded clause syntaxes over Π_m . Then $Q_m \leq c_{\text{cl}} \sum_{\ell=0}^t |\Pi_m|^{\ell+1} \leq C_{\text{cl}}(m+1)^{t+1}$ for a fixed constant C_{cl} .

3.3 Candidate-rejection entropy upper bounds

Proposition 3.4 (Universal entropy upper bound). *Fix $k \geq 2$ and a bounded FICS regime $\mathfrak{B}$. There exists a constant $C_{\mathfrak{B},k} > 0$ such that*

$$|\mathcal{V}_k^{\mathfrak{B}}(m)| \leq 2^{C_{\mathfrak{B},k} m \log_2(m+1)}.$$

Equivalently,

$$\log_2 |\mathcal{V}_k^{\mathfrak{B}}(m)| = O_{\mathfrak{B},k}(m \log(m+1)).$$

Proof. For a fixed regime $\mathfrak{B}$, the number of canonical systems with at most m clauses is at most $(Q_m + 1)^m$. The choice of a start predicate and of an admissible candidate contributes only polynomial factors depending on the fixed regime. Thus $|\mathcal{V}_k^{\mathfrak{B}}(m)| \leq |\Pi_m| Q_m (Q_m + 1)^m$. Taking logarithms and using $|\Pi_m| = O_{\mathfrak{B}}(m)$ and $Q_m = O_{\mathfrak{B}}((m+1)^{t+1})$ gives the result. $\square$

Theorem 3.5 (Universal fiber-profile upper bound). *Fix $k \geq 2$ and a bounded FICS regime $\mathfrak{B}$. For every $1 \leq r \leq k$,*

$$d_r^{\mathfrak{B}}(m) = O_{\mathfrak{B},k,r}((m \log(m+1))^{1/r}).$$

Proof. A fiber class contains no more concepts than $\mathcal{V}_k^{\mathfrak{B}}(m)$. If it shatters an r -box of side D , then it realizes all 2^{D^r} labelings, so $2^{D^r} \leq |\mathcal{V}_k^{\mathfrak{B}}(m)|$. Proposition 3.4 yields $D^r = O_{\mathfrak{B},k}(m \log(m+1))$, which proves the claim. $\square$

3.4 Target-induced fixed-target candidate capacity

The preceding bounds allow the target FICS itself to vary. We now fix one target presentation Γ and vary only admissible candidates over its active predicate universe $\text{Pred}^*(\Gamma)$. The resulting target-induced candidate class is much smaller. Define $\mathcal{V}_k(\Gamma) = \{V_{\Gamma,\gamma} : \gamma \text{ is an admissible } k\text{-label candidate over } \Gamma\}$. For $1 \leq r \leq k$, define $d_r(\Gamma)$ from $\mathcal{V}_k(\Gamma)$ by the same fiber operation as above.

Proposition 3.6 (Fixed-target entropy and fiber bound). *Fix $k \geq 2$ and a bounded regime $\mathfrak{B}$. There is a constant $C_{\mathfrak{B},k}$ such that every FICS $\Gamma \in \mathfrak{B}$ with at most m clauses satisfies*

$$|\mathcal{V}_k(\Gamma)| \leq C_{\mathfrak{B},k}(m+1)^{t+1}$$

and hence

$$\log_2 |\mathcal{V}_k(\Gamma)| = O_{\mathfrak{B},k}(\log(m+1)).$$

Consequently, for every $1 \leq r \leq k$,

$$d_r(\Gamma) = O_{\mathfrak{B},k,r}((\log(m+1))^{1/r}).$$

The same bounds hold for the live-candidate subclass over Γ .

Proof. By Lemma 3.1, or directly by counting predicate occurrences, $|\text{Pred}^*(\Gamma)| = O_{\mathfrak{B}}(m+1)$. Once the target is fixed, each candidate is one bounded clause syntax over this fixed active predicate pool. The bounded-clause count gives at most $C_{\mathfrak{B},k}(m+1)^{t+1}$ admissible candidates and therefore no more candidate-rejection relations. If an r -fiber shatters a box of side D , then $2^{D^r} \leq |\mathcal{V}_k(\Gamma)|$, so $D^r = O_{\mathfrak{B},k}(\log(m+1))$. $\square$

Remark 3.7 (Target-induced versus learner-state capacity). The global $m \log m$ entropy scale and the fixed-target $\log m$ scale answer different target-relative questions. In the global class, both the target presentation and the admissible target-relative candidate may vary. In the fixed-target class, the target is fixed and only admissible candidates over $\text{Pred}^*(\Gamma)$ vary.

Neither statement is a capacity bound for the full sample-induced candidate space constructed by Algorithm 3 over a current learner basis F . Such a basis may contain boundary representatives having no corresponding predicate in the target presentation. Our fixed-target class instead isolates the subfamily induced by target predicates; for live predicates, this subfamily is embedded into the learner's candidate space through representative translation once the required representatives have been exposed.

In particular, the global shattering lower bound does not say that one execution on one fixed target realizes all labelings of the extremal box.

4 Sharp Lower Bounds via FICS Realization

The lower bound does not rely on a new VC estimate for finite automata. Its finite-state component is representational: fixed-length residual compression gives a compact code for an arbitrary box labeling, and an FCP-preserving linear-size simulation transfers that code into bounded rank-two FICS syntax.

The technical work is therefore a transfer problem rather than a finite-state VC calculation. The transfer must simultaneously satisfy seven constraints: the labeling is compressed into fixed-length DFA residuals; the automaton is simulated with only linear syntactic overhead; the simulation preserves FCP; all relevant predicates remain live; an exact length gate restricts rejection to the intended finite box; anchored FCP contexts and ordered-boundary

positive examples expose the predicates and block residuals needed by the learner; and the resulting translated clause must occur in the exact bounded candidate syntax searched by Algorithm 3. The fixed-target construction adds an eighth requirement: all classifiers for all fiber dimensions must coexist inside one master target.

The global construction is organized around fixed-length residual compression, an FCP-preserving simulation of partial DFAs by rank-two FICSs, and an r -active/ $(k - r)$ -dummy separator gadget.

4.1 Fixed-length residual compression

Lemma 4.1 (Fixed-length residual compression). *For every $L \geq 8$ and every $F \subseteq \{0, 1\}^L$, there is an accessible partial deterministic finite automaton recognizing exactly F and having fewer than $12 \frac{2^L}{L}$ states.*

Proof. The empty language is trivial, so assume $F \neq \emptyset$. For a prefix $v \in \{0, 1\}^i$, define $R_v = \{z \in \{0, 1\}^{L-i} : vz \in F\}$. At level i , merge prefixes with the same nonempty residual, retaining the level as part of the state. Transitions are well defined because $R_v = R_w$ implies $R_{va} = R_{wa}$ for $a \in \{0, 1\}$. The only nonempty residual at level L is $\{\varepsilon\}$, which is accepting.

The number of states at level i is at most both 2^i and $2^{2^{L-i}} - 1$. Hence $|Q| \leq \sum_{i=0}^L \min\{2^i, 2^{2^{L-i}} - 1\}$. Put $j = L - i$ and $J = \max\{j : 2^j \leq L - j\}$. Then $|Q| \leq \sum_{j=0}^J 2^{2^j} + \sum_{j=J+1}^L 2^{L-j}$. The first sum is at most $2 \cdot 2^{2^J}$, and the second is less than 2^{L-J} . Since $2^J \leq L - J$, $|Q| < 3 \cdot 2^{L-J}$. By maximality of J , $2^{J+1} > L - J - 1$. For $L \geq 8$, $J + 1 \leq L/2$, hence $2^J > L/4$. Therefore $|Q| < 12 \frac{2^L}{L}$. $\square$

Remark 4.2. The $n \log n$ capacity scale for finite automata is classical; see Ishigami and Tani [7]. The role of finite automata here is not to supply a new finite-state VC theorem, but to provide an explicit fixed-length encoding of arbitrary box labelings that can be transferred into FICS syntax with linear overhead and FCP preservation. The self-contained lemma above also avoids dependence on complete-versus-partial DFA conventions.

4.2 Rank-two path encoding

Fix finite vertex and edge alphabets containing fresh labels $\{\mathbf{N}, \mathbf{L}, \mathbf{R}\}$ and $\{0, 1, \#, \mathbf{b}, @, \$\}$, respectively. The marker vertex labels $\mathbf{L}, \mathbf{R}$ and marker edge labels $@, \$$ never occur in internal word paths.

For a nonempty word $u = a_1 \cdots a_\ell$, define the rank-two ground graph pattern $P[u]$ by

$$V(P[u]) = \{v_0, \dots, v_\ell\}, \quad E(P[u]) = \{\{v_{i-1}, v_i\} : 1 \leq i \leq \ell\},$$

with all vertex labels equal to $\mathbf{N}$, edge $\{v_{i-1}, v_i\}$ labeled a_i , and ordered interface $\iota(P[u]) = (v_0, v_\ell)$. Although the ordinary edges are undirected, the ordered interface records the direction of the encoded word.

For a rank-two variable x , let S_x be the star pattern with two neutral vertices s_0, s_1 , one variable hyperedge labeled x with ordered ports (s_0, s_1) , no ordinary edges, and interface (s_0, s_1) . Thus $S_x\{x := K\} \cong_\iota K$ for every rank-two ground graph pattern K .

For each ordinary edge label a , define the prefix frame $E_a[x]$ on neutral vertices e_0, e_1, e_2 , with one ordinary a -edge $\{e_0, e_1\}$, one variable hyperedge x with ports (e_1, e_2) , and interface (e_0, e_2) . By the binding operation,

$$\boxed{\text{Real}(E_a[x], x := P[u]) \cong_\iota P[au].} \tag{1}$$

Define the rank-zero wrapper $W[x]$ on vertices ℓ, z_0, z_1, r , with labels $\phi(\ell) = \mathbf{L}$, $\phi(z_0) = \phi(z_1) = \mathbf{N}$, $\phi(r) = \mathbf{R}$, ordinary edges $\{\ell, z_0\}$ and $\{z_1, r\}$ labeled $@$ and $\$$, respectively, one

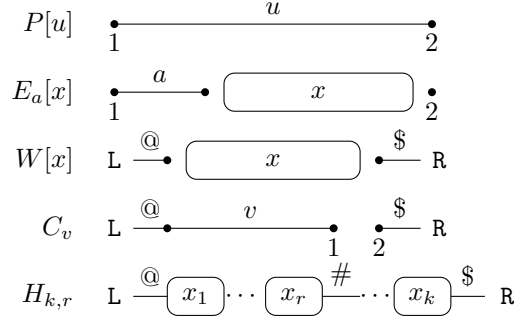


Figure 1: Schematic lower-bound gadgets. Boxes denote variable hyperedges; numbered endpoints indicate ordered interfaces. Ordinary graph edges are undirected; orientation of encoded words is carried by interface order.

variable hyperedge x with ports (z_0, z_1) , and empty interface. Put $\widehat{P}[u] = \text{Real}(W[x], x := P[u])$.

4.3 DFA simulation inside exact FICS syntax

Let $A = (Q, \Sigma, \delta, q_0, F)$ be an accessible partial DFA. Introduce a rank-two predicate Q_q for each state $q \in Q$. For every defined transition $\delta(q, a) = q'$, add

$$Q_q(E_a[x]) \leftarrow Q_{q'}(S_x). \quad (2)$$

Whenever $\delta(q, a) \in F$, add the fact

$$Q_q(P[a]) \leftarrow . \quad (3)$$

Finally introduce a rank-zero start predicate p and

$$p(W[x]) \leftarrow Q_{q_0}(S_x). \quad (4)$$

Lemma 4.3 (Syntactic validity of the automaton clauses). *Every clause in Γ_A is a fixed-interface clause in the sense of the FICS syntax. In (2) and (4), the unique head variable-hyperedge occurrence is paired with the unique body star atom carrying the same variable label and rank; in (3) both sets are empty.*

Proof. This is immediate from the explicit definitions of $E_a[x]$, S_x , $P[a]$, and $W[x]$. The head/body variable-occurrence correspondence is bijective and preserves the ordered rank-two interface. $\square$

Lemma 4.4 (DFA simulation). *For every state q and every rank-two ground graph pattern K , $\Gamma_A \models Q_q(K)$ iff there is a nonempty word u such that $K \cong_\iota P[u]$ and $\delta(q, u) \in F$.*

Proof. Induct on the length of a refutation. A fact applies exactly to a one-edge path $P[a]$ with $\delta(q, a) \in F$. Otherwise the first clause must be a transition clause, forcing $K \cong_\iota \text{Real}(E_a[x], x := K')$ with $\Gamma_A \models Q_{\delta(q,a)}(K')$. The induction hypothesis gives $K' \cong_\iota P[v]$. By (1), $K \cong_\iota P[av]$, and the automaton condition follows. The converse is the same induction read along the word. $\square$

Consequently, $\Gamma_A \models p(G)$ iff $G \cong \widehat{P}[u]$ for a nonempty word u accepted from q_0 .

4.4 Anchored contexts and FCP

For a word $v = a_1 \cdots a_j$, define a rank-two context C_v on distinct vertices $\ell, c_0, \dots, c_j, y, r$. The vertices ℓ, r have labels L, R, respectively, and all others have label N. The ordinary edges are $\{\ell, c_0\}$, labeled @, the path edges $\{c_{i-1}, c_i\}$, labeled a_i , and $\{y, r\}$, labeled \$. Its ordered interface is $\iota(C_v) = (c_j, y)$. In particular, C_v contains no ordinary edge $\{c_j, y\}$. Whenever the composition is defined,

$$C_v \odot P[u] \cong \widehat{P}[vu]. \quad (5)$$

Lemma 4.5 (No cross-component edge collision). *Let G and K be rank-two graphs with ordered interfaces $\iota(G) = (g_1, g_2)$, $\iota(K) = (k_1, k_2)$, and suppose $G \odot K$ is defined in the sense of [14]. An ordinary edge of G and an ordinary edge of K can have the same image in the glued graph only if they are the respective edges joining the two interface vertices. Consequently, if G has no ordinary edge $\{g_1, g_2\}$, then every ordinary edge of K has a distinct surviving image in $G \odot K$, and every non-interface vertex of K survives as a distinct vertex.*

Proof. Before gluing, take disjoint copies of G and K . The quotient identifies only g_1 with k_1 and g_2 with k_2 ; no non-interface vertex of one component is identified with a vertex of the other component. Hence an edge $\{u, v\}$ of G and an edge $\{u', v'\}$ of K can acquire the same unordered pair of endpoints only when both endpoints of each edge belong to the respective interfaces. Since the interfaces have rank two, this forces the two edges to be $\{g_1, g_2\}$ and $\{k_1, k_2\}$, respectively. The final assertions are immediate. $\square$

Lemma 4.6 (Anchored suffix rigidity). *Suppose $C_v \odot K$ is defined and $C_v \odot K \cong \widehat{P}[w]$. Then there is a unique nonempty word u such that $w = vu$ and $K \cong_\iota P[u]$.*

Proof. The marked path $\widehat{P}[w]$ contains exactly one vertex labeled L, exactly one vertex labeled R, and unique incident marker edges labeled @ and \$. Any isomorphism therefore fixes the two marker ends and forces the entire context prefix $\ell, c_0, \dots, c_j$ to coincide with the initial segment labeled v . Hence v is a prefix of w ; write $w = vu$.

The two components in $C_v \odot K$ share only the two ordered interface vertices. The marker labels L, R, @, \$ are disjoint from the internal word labels, so no internal feature of K can be confused with either external marker. Since C_v has no ordinary edge between its two interface vertices, Lemma 4.5 shows that every ordinary edge of K survives distinctly in the composition and every non-interface vertex of K survives as a distinct vertex. Any branch, chord, cycle, extra edge, or extra non-interface vertex in K would therefore survive and make the result non-isomorphic to the simple marked path $\widehat{P}[w]$. Consequently K is exactly the remaining suffix path, with its first and second interface vertices mapped to the left and right ends of that suffix. Hence $K \cong_\iota P[u]$. Uniqueness follows from the uniqueness of the suffix after a fixed prefix. $\square$

4.5 Boundary exposure of anchored path residuals

The learner of [14] constructs its residual set from valid ordered boundary specifications in positive examples. For the anchored path contexts above, the inserted suffix path is recovered by one explicit rank-two specification.

Lemma 4.7 (Explicit exposure of an anchored path residual). *Let v be a word and let $u = a_1 \cdots a_h$ with $h \geq 1$. Put $G_{v,u} = C_v \odot P[u] \cong \widehat{P}[vu]$. Let $t_0, t_1, \dots, t_h$ be the vertices of the copy of $P[u]$ inside $G_{v,u}$, in path order, where t_0 and t_h are the two vertices identified with the ordered interface of C_v . Define $\beta_u = (t_0, t_h)$ and let $E_u^\partial = \{e \in E(P[u]) : e \cap \{t_0, t_h\} \neq \emptyset\}$,*

where the edges are understood as their copies in $G_{v,u}$. Then (β_u, E_u^∂) is a valid ordered boundary specification in the sense of Definition 2.5 of [14], and

$$\boxed{K_{G_{v,u}}(\beta_u, E_u^\partial) \cong_\iota P[u].}$$

Consequently, if $G_{v,u}$ occurs in a positive sample D_n and $w \geq 2$, then $\text{BRep}_w(D_n)$ contains a representation $\sigma_{v,u}$ such that $\text{can}(\sigma_{v,u}) \cong_\iota P[u]$.

Proof. The only vertices shared by the anchored context C_v and the inserted path $P[u]$ are t_0 and t_h . Every edge in E_u^∂ is incident with one of these boundary vertices, so (β_u, E_u^∂) is valid.

Suppose first that $h \geq 2$. After deleting t_0 and t_h from $G_{v,u}$, the vertices $t_1, \dots, t_{h-1}$ form one connected path component. The non-boundary endpoints of the edges in E_u^∂ lie in this component, so the reachable set $I_{G_{v,u}}(\beta_u, E_u^\partial)$ of Definition 2.5 is exactly $\{t_1, \dots, t_{h-1}\}$. The context edge immediately preceding t_0 and the marker edge incident with t_h are not in E_u^∂ ; after deletion of the boundary vertices, their remaining context components are therefore not reachable from the selected boundary edges. Hence $K_{G_{v,u}}(\beta_u, E_u^\partial)$ contains precisely the vertices $t_0, \dots, t_h$ and precisely the u -labeled path edges between them.

If $h = 1$, there are no interior vertices and E_u^∂ consists of the single edge joining the two boundary vertices, so the same conclusion is immediate. The boundary order (t_0, t_h) agrees with the ordered interface of $P[u]$. Therefore $K_{G_{v,u}}(\beta_u, E_u^\partial) \cong_\iota P[u]$.

Algorithm 1 of [14] enumerates all valid ordered boundary specifications of rank at most w arising from the positive graphs in D_n . Since the present specification has rank two, it is included in $\text{BRep}_w(D_n)$ whenever $w \geq 2$, and its canonical graph is interface-isomorphic to $P[u]$. $\square$

Lemma 4.8 (Explicit exposure of an anchored context). *With the notation of Lemma 4.7, let $E_v^\partial = \{e \in E(C_v) : e \cap \{t_0, t_h\} \neq \emptyset\}$, where the edges are understood as their copies in $G_{v,u}$. Then (β_u, E_v^∂) is a valid rank-two ordered boundary specification and*

$$\boxed{K_{G_{v,u}}(\beta_u, E_v^\partial) \cong_\iota C_v.}$$

Consequently, if $G_{v,u}$ occurs in a positive sample D_n and $w \geq 2$, then $\text{BRep}_w(D_n)$ contains a representation $\rho_{v,u}$ with $\text{can}(\rho_{v,u}) \cong_\iota C_v$.

Proof. This is the same boundary-separation argument used in the proof of Lemma 4.15 of [14]. After deleting the two shared interface vertices t_0, t_h , the non-interface part of the copy of C_v is disjoint from the non-interface part of the inserted path $P[u]$. The boundary edge set E_v^∂ selects precisely the components on the C_v -side of the cut. Definition 2.5 therefore reconstructs exactly the canonical copy of C_v , with the original interface order. Algorithm 1 then includes this rank-two representation in $\text{BRep}_w(D_n)$. $\square$

Let $\mathbf{0}$ denote the empty rank-zero ground graph pattern. Whenever rank-zero composition is taken with $\mathbf{0}$, it leaves the other rank-zero ground graph unchanged up to isomorphism.

Proposition 4.9 (FCP-preserving automaton embedding). *If A is accessible, then (Γ_A, p) satisfies FCP in the sense of [14]. Moreover, $|\Gamma_A| = O(|Q|)$, with fixed finite alphabets, rank-two auxiliary predicates, bounded degree, and a fixed finite family of local head frames.*

Proof. For $q \in Q$, choose an access word v_q with $\delta(q_0, v_q) = q$. Consider any rank-two ground graph pattern K for which $C_{v_q} \odot K$ is defined.

If $\Gamma_A \models Q_q(K)$, Lemma 4.4 gives $K \cong_\iota P[u]$ and $\delta(q, u) \in F$. Hence, by (5), $C_{v_q} \odot K \cong \hat{P}[v_q u]$ and the start predicate accepts the composition.

Conversely, if $\Gamma_A \models p(C_{v_q} \odot K)$, the start-language characterization gives $C_{v_q} \odot K \cong \widehat{P}[w]$ for an accepted word w . Lemma 4.6 yields $w = v_q u$ and $K \cong_i P[u]$. Therefore $\delta(q, u) = \delta(q_0, v_q u) \in F$, and Lemma 4.4 gives $\Gamma_A \models Q_q(K)$. Thus C_{v_q} witnesses FCP for Q_q . The start predicate is witnessed by $\mathbf{0}$, since $\mathbf{0} \odot G \cong G$ for every rank-zero ground graph G .

There are at most $|Q||\Sigma|$ transition clauses and at most $|Q||\Sigma|$ terminal facts, plus one wrapper clause. Since Σ is fixed, the clause count is $O(|Q|)$. All graph frames used above belong to a fixed finite family independent of $|Q|$. $\square$

4.6 r -active realization with a length gate

Fix $1 \leq r \leq k$, $h \geq 1$, and

$$D = 2^h, \quad A_D = \{[P[u]]_{\cong_i} : u \in \{0, 1\}^h\}.$$

Because ordered-interface isomorphism preserves the edge-label sequence, $|A_D| = D$.

Let $S \subseteq (\{0, 1\}^h)^r$ be arbitrary and define

$$F_S = \{u_1 \cdots u_r : (u_1, \dots, u_r) \notin S\} \subseteq \{0, 1\}^{rh}.$$

When $F_S \neq \emptyset$, apply the residual-compression lemma to obtain a partial DFA with $O_r(D^r / \log D)$ states. Extend its classification branch so that it recognizes $F_S \# \{0, 1\}^*$: the fixed-length final residual is made nonaccepting and receives a $\#$ -transition to an accepting state T , and T loops on 0, 1. If $F_S = \emptyset$, use a constant-size rejecting classification branch; if $F_S = \{0, 1\}^{rh}$, use the ordinary residual construction before the separator extension.

To make the body support exact, add a fresh access symbol $\mathbf{b}$ and a length-gate chain $B_0, B_1, \dots, B_h$ with

$$\delta(q_0, \mathbf{b}) = B_0, \quad \delta(B_j, a) = B_{j+1} \quad (0 \leq j < h, a \in \{0, 1\}),$$

where B_h is accepting and has no outgoing 0, 1-transition. The gate is disjoint from the classification branch. Hence

$$\boxed{\Gamma \models Q_{B_0}(P[u]) \iff |u| = h.} \quad (6)$$

The chain adds only $O(h) = O(\log D)$ states, which is negligible relative to $O_r(D^r / \log D)$ for the asymptotic lower bound. The symbol $\mathbf{b}$ never occurs in candidate-head strings.

For $r < k$, define the rank-zero head scaffold $H_{k,r}[x_1, \dots, x_k]$ using neutral vertices $z_0, \dots, z_r, z_r^+, z_{r+1}, \dots, z_k$ and k variable hyperedges. For $1 \leq i \leq r$, the ports of x_i are (z_{i-1}, z_i) ; the ports of x_{r+1} are (z_r^+, z_{r+1}) ; and subsequent variables continue in series. Add the ordinary separator edge $\{z_r, z_r^+\}$ labeled $\#$, together with the external L/@ and R/\$ markers. For $r = k$, place the $\#$ -edge after the final variable slot and before the right marker. Then

$$\text{Real}(H_{k,r}; x_i := P[u_i]) \cong \widehat{P}[u_1 \cdots u_r \# u_{r+1} \cdots u_k] \quad (7)$$

for $r < k$, with the evident terminal-separator version when $r = k$.

Define the live candidate

$$\gamma_{k,r}^{(h)} : p(H_{k,r}[x_1, \dots, x_k]) \leftarrow Q_{B_0}(S_{x_1}), \dots, Q_{B_0}(S_{x_k}). \quad (8)$$

Both p and Q_{B_0} are live: the length-gate branch itself supplies accepted start graphs, and Q_{B_0} accepts every length- h binary path.

Lemma 4.10 (Exact fixed-interface validity of the lower-bound candidate). *The candidate $\gamma_{k,r}^{(h)}$ is a fixed-interface clause with exactly k distinct rank-two variable labels. Its k head variable-hyperedge occurrences are in bijection with the k body star atoms, and the corresponding occurrence and star carry the same label and ordered rank-two interface.*

Proof. Each variable hyperedge x_i occurs exactly once in $H_{k,r}$. The body contains exactly one star S_{x_i} , which has no ordinary edge, one variable hyperedge labeled x_i , and interface equal to its ordered ports. Thus the occurrence-to-body-star correspondence is $h_i \longleftrightarrow Q_{B_0}(S_{x_i})$, $1 \leq i \leq k$, a bijection preserving label and rank. This is precisely the occurrence-based fixed-interface condition; distinct substitution labels index the product arity separately. $\square$

For $S \subseteq (\{0, 1\}^h)^r$, write $S^\# = \{(P[u_1], \dots, P[u_r]) : (u_1, \dots, u_r) \in S\}$.

Lemma 4.11 (Exact support of the lower-bound relation). *For the length-gated construction,*

$$V_{\Gamma_{D,S}^{(r)}, \gamma_{k,r}^{(h)}} = S^\# \times A_D^{k-r}.$$

In particular the rejection relation vanishes outside A_D^k .

Proof. If some coordinate K_i is not a length- h binary path in A_D , the corresponding body atom $Q_{B_0}(S_{x_i})$ is not refutable, by the exact length gate, so the tuple is not rejected. If every coordinate is $P[u_i]$ with $|u_i| = h$, all body atoms are refutable. By the separator construction in (7), the head belongs to the classification language exactly when $u_1 \cdots u_r \in F_S$, i.e. exactly when $(u_1, \dots, u_r) \notin S$. The body-positive/head-negative convention therefore labels precisely the tuples in $S^\# \times A_D^{k-r}$. $\square$

Lemma 4.12 (Context compatibility of the extremal box). *For every lower-bound target/candidate pair $(\Gamma_{D,S}^{(r)}, \gamma_{k,r}^{(h)})$, choose C_b as the FCP context of the length-gate predicate Q_{B_0} , and choose the empty rank-zero context $\mathbf{0}$ for the start predicate. Then*

$$A_D^k \subseteq \Omega_{\Gamma_{D,S}^{(r)}, \gamma_{k,r}^{(h)}, C}.$$

Proof. Take $(K_1, \dots, K_k) \in A_D^k$, so $K_i = P[u_i]$ with $|u_i| = h$. Each body star realizes to $P[u_i]$. The access-word construction for the gate state gives the context C_b ; its interface labels agree with those of $P[u_i]$, and it has no ordinary edge joining its two interface vertices. Hence $C_b \odot P[u_i]$ is defined and equals the marked path $\hat{P}[bu_i]$ up to isomorphism. The realized head has rank zero, and composition with the empty rank-zero context $\mathbf{0}$ is always defined. Thus all realized body and head arguments satisfy the compatibility conditions in the definition of Ω . $\square$

Theorem 4.13 (Uniform rich-regime realization). *For every fixed $k \geq 2$, there exists one bounded FICS regime $\mathfrak{B}_k$ such that, simultaneously for every $1 \leq r \leq k$, all sufficiently large powers of two $D = 2^h$, and every $S \subseteq (\{0, 1\}^h)^r$ (equivalently, every $S^\# \subseteq A_D^r$), there exist an FCP-FICS $\Gamma_{D,S}^{(r)} \in \mathfrak{B}_k$ and a live admissible candidate $\gamma_{k,r}^{(h)}$ with exactly k distinct rank-two variable labels satisfying*

$$|\Gamma_{D,S}^{(r)}| \leq C_{k,r} \frac{D^r}{\log D},$$

and

$$V_{\Gamma_{D,S}^{(r)}, \gamma_{k,r}^{(h)}} = S^\# \times A_D^{k-r}.$$

More explicitly, one may take

$$\Delta = 2, \quad s = t = k, \quad w = 2,$$

choose a constant $d = d_k$, and fix one finite admissible head-frame family $\mathcal{H}_{d_k}$ containing every frame type used by the construction. The choices d_k , $\Sigma_{V,k}$, $\Sigma_{E,k}$, and $\mathcal{H}_{d_k}$ depend only on k , not on r , D , S , or m .

Proof. The residual classifier uses $O_r(D^r / \log D)$ states and the length gate adds only $O(\log D)$ states. Proposition 4.9 gives linear clause overhead. For fixed k , let $\mathcal{H}_{d_k}$ contain the finitely many head-frame types used by the transition clauses, terminal facts, wrapper and gate clauses, together with the candidate scaffolds $H_{k,1}, \dots, H_{k,k}$. Choose d_k large enough to bound the vertex degree of every one of these finitely many local frames, take $s = t = k$ and $w = 2$, and fix the finite alphabets used in the construction. Every graph generated by the start predicate is an externally marked path of maximum ordinary degree two, so the semantic degree bound may be fixed as $\Delta = 2$. Hence every target and candidate used here satisfies the same bounded parameters of Definition 2.25 of [14], independently of r, D, S, m , and all targets lie in one fixed presentation-level subregime $\mathfrak{B}_k$ of $\mathcal{FICS}\mathcal{L}_2^{\text{FCP}}(m, k, k, 2, d_k)$. Lemma 4.11 gives the claimed exact relation. $\square$

Remark 4.14 (Dependence of hidden constants). The present theorem fixes k and r ; no uniformity in growing k is claimed. The finite-state part of the lower bound is explicit. Since the residual word length is $L = rh$ and $D = 2^h$, the residual-compression lemma uses fewer than $12 \frac{D^r}{r \log_2 D}$ states before the constant-size separator extension and the $h + 1$ -state length gate. The remaining lower-bound constants come from the fixed linear DFA-to-FICS simulation and the finite k -dependent head-frame family. The upper-bound constants additionally depend on the number of admissible bounded clause skeletons and variable-equality patterns in the chosen regime. Determining useful uniform dependence on growing k is left to the joint (m, k) problem in Section 5.

Theorem 4.15 (Learner realization of the extremal box). *For every target/candidate pair supplied by Theorem 4.13, let $C_{\mathbf{b}}$ be the FCP context for the accessible gate state B_0 , obtained from the access word $\mathbf{b}$. Consider a learner stage n at which the current predicate basis F contains a representative ρ_B with $\text{can}(\rho_B) = C_{\mathbf{b}}$. The special rank-zero representation $\rho_{\emptyset}$ is available to Algorithm 3 by definition.*

Under every complete positive presentation, there is an integer N_D such that for every $n \geq N_D$ the current residual set $R = \text{BRep}_2(D_n)$ contains representations σ_u , one for every $u \in \{0, 1\}^h$, satisfying $\text{can}(\sigma_u) \cong_i P[u]$. The same positive anchored examples also expose a representation of $C_{\mathbf{b}}$; hence every update stage $n \geq N_D$ that occurs in the learner run has a suitable $\rho_B \in F$. Consequently, at every such update stage the representative translation

$$\tilde{\gamma}_{k,r}^{(h)} : p_{\rho_{\emptyset}}(H_{k,r}) \leftarrow p_{\rho_B}(S_{x_1}), \dots, p_{\rho_B}(S_{x_k})$$

belongs to the bounded candidate set of Algorithm 3, and its Cartesian-product search explicitly examines every tuple in A_D^k . The resulting body-positive/head-negative rejection bits on that box are exactly $S^{\sharp} \times A_D^{k-r}$. For each labeling S there is such a target/candidate pair; the theorem does not assert that one fixed target or one learner run realizes all labelings. This is a conditional realization statement: it asserts the behavior of every update stage occurring after N_D , but does not assert the existence of such an update stage.

Proof. The gate state B_0 is accessible by $\mathbf{b}$, so the anchored-context construction used in Proposition 4.9 gives its FCP context $C_{\mathbf{b}}$. For every $u \in \{0, 1\}^h$, the exact length gate gives $\Gamma_{D,S}^{(r)} \models Q_{B_0}(P[u])$, and hence FCP gives the positive start-language graph $G_u = C_{\mathbf{b}} \odot P[u] \cong \hat{P}[\mathbf{b}u]$. Because the set of these graphs is finite, a complete positive presentation contains all of them from some stage N_D onward. Algorithm 4 of [14] recomputes $R = \text{BRep}_2(D_n)$ at every stage in the present regime. Lemma 4.7 therefore supplies, for every $n \geq N_D$ and every $u \in \{0, 1\}^h$, a residual representation $\sigma_u \in R$ with $\text{can}(\sigma_u) \cong_i P[u]$. Applying Lemma 4.8 to any one of the same positive graphs gives a representation of $C_{\mathbf{b}}$ in $\text{BRep}_2(D_n)$. At an update stage Algorithm 4 resets $F = \text{BRep}_2(D_n)$, so a suitable ρ_B then belongs to F .

At every update stage $n \geq N_D$ that occurs in the learner run, Theorem 2.6 and Lemma 4.10 show directly from the bounded-candidate construction of Algorithm 3 that $\tilde{\gamma}_{k,r}^{(h)}$ is one of the enumerated candidates. The Cartesian-product loop of Algorithm 3 ranges over all families in $R^{X(\gamma)}$, so the representations σ_u ensure that every canonical tuple in A_D^k is explicitly tested. By Lemma 4.12, the entire box lies in the context-compatible domain, and Theorem 2.6 identifies the learner-side rejection bit with $V_{\Gamma,\gamma}$ pointwise there. Finally, Lemma 4.11 gives exactly $S^\# \times A_D^{k-r}$. $\square$

Corollary 4.16 (Spuriousness of the extremal candidate). *For the length-gated lower-bound pair, $V_{\Gamma_{D,S},\gamma_{k,r}}^{(r)} = \emptyset \iff S = \emptyset$. Thus, after representative and witness exposure, a nonempty S supplies exactly the substitution geometry by which the corresponding learner candidate is rejected as spurious.*

4.7 Sharp fiber profile and entropy

Theorem 4.17 (Sharp live and full profiles in one fixed rich regime). *For every fixed $k \geq 2$, the regime $\mathfrak{B}_k$ of Theorem 4.13 satisfies, simultaneously for every $1 \leq r \leq k$,*

$$d_{r,\text{live}}^{\mathfrak{B}_k}(m) = \Theta_{k,r}\left((m \log(m+1))^{1/r}\right)$$

and therefore also

$$d_r^{\mathfrak{B}_k}(m) = \Theta_{k,r}\left((m \log(m+1))^{1/r}\right).$$

In particular,

$$\text{VC}_k^\square(\mathcal{V}_k^{\mathfrak{B}_k}(m)) = \Theta_k\left((m \log(m+1))^{1/k}\right)$$

and

$$d_1^{\mathfrak{B}_k}(m) = \Theta_k(m \log(m+1)).$$

Proof. The upper bound is Theorem 3.5, applied to the fixed regime $\mathfrak{B}_k$. For the lower bound, Theorem 4.13 uses live candidates and shatters an r -box of side D using at most $C_{k,r}D^r / \log D$ clauses. Choosing a power of two $D = \Theta_{k,r}((m \log(m+1))^{1/r})$ with a sufficiently small implicit constant keeps the clause budget at most m , yielding the matching lower bound. $\square$

Corollary 4.18 (Sharp candidate-rejection entropy in the rich regime). *For the same regime $\mathfrak{B}_k$,*

$$\log_2 |\mathcal{V}_{k,\text{live}}^{\mathfrak{B}_k}(m)| = \Theta_k(m \log(m+1))$$

and

$$\log_2 |\mathcal{V}_k^{\mathfrak{B}_k}(m)| = \Theta_k(m \log(m+1)).$$

Consequently, for every $1 \leq r \leq k$,

$$\log_2 |\mathcal{V}_k^{\mathfrak{B}_k}(m)| = \Theta_{k,r}((d_r^{\mathfrak{B}_k}(m))^r).$$

Proof. The upper bound is Proposition 3.4. For the lower bound, take $r = k$ in the live lower bound of Theorem 4.17: a box of side $\Omega_k((m \log(m+1))^{1/k})$ is shattered, hence the live concept class contains $2^{\Omega_k(m \log(m+1))}$ distinct restrictions. This gives the live lower bound, while the full-class lower bound follows by inclusion; the universal entropy upper bound applies to both classes. $\square$

Corollary 4.19 (Unified information scale). *For every fixed $1 \leq r \leq k$, in the rich regime $\mathfrak{B}_k$,*

$$\boxed{(d_{r,\text{live}}^{\mathfrak{B}_k}(m))^r \asymp_{k,r} \log_2 |\mathcal{V}_{k,\text{live}}^{\mathfrak{B}_k}(m)| \asymp_k m \log(m+1),}$$

and the same asymptotic scale holds for the full candidate class.

Remark 4.20. The word *simultaneously* refers to the common ambient regime $\mathfrak{B}_k$. The extremal grammar-candidate pair need not be the same for different r . Our lower bound uses r active variables and $k - r$ semantically dummy variables. Moreover, shattering varies the target FICS with the labeling. Thus the theorem is a class-level capacity statement, not a claim that one fixed target or one learner run realizes every labeling.

4.8 Simultaneously sharp fixed-target profiles

We now prove the fixed-target half of the two-level capacity theorem. Unlike the global lower bound, this construction is not allowed to change the target when the desired labeling changes. Instead, one master FICS stores, inside its predicate structure, a finite *bank* of live classifiers for every fiber dimension $1 \leq r \leq k$. The candidate syntax then selects one classifier from this bank through its head predicate. This realizes every labeling needed for the relevant shattered box while keeping the target fixed.

The construction is the mechanism behind the separation $m \log m$ versus $\log m$: the former allows the target presentation itself to change with the labeling, whereas the latter forces all labeling information to coexist inside one bounded target.

Fix $k \geq 2$. Let n be a sufficiently large integer and put $L = \lfloor \log_2 n \rfloor$. For $1 \leq r \leq k$, define

$$h_r = \left\lfloor \frac{L}{r} \right\rfloor, \quad D_r = 2^{h_r}, \quad N_r = D_r^r = 2^{r h_r},$$

and $A_r = \{[P[u]]_{\cong_\ell} : u \in \{0, 1\}^{h_r}\}$. For $n \geq 2^k$, every h_r is positive. Writing $L = qr + s$ with $0 \leq s < r$ gives $N_r = 2^{L-s}$, and therefore

$$\frac{n}{2^r} < 2^{L-r+1} \leq N_r \leq 2^L \leq n. \quad (9)$$

Thus $N_r = \Theta_{k,r}(n)$ simultaneously for all $1 \leq r \leq k$.

A master accessible DFA. Use the fixed alphabet $\Sigma^* = \{0, 1, \#, \lambda, \alpha, \beta\}$. The symbols $0, 1, \#$ are used by candidate-head words, λ is a private liveness symbol, and α, β are used only for selector prefixes.

For each r and each $S \subseteq (\{0, 1\}^{h_r})^r$, create a classifier root $q_{r,S}$. From $q_{r,S}$, place a complete binary trie of depth $r h_r$. Thus the state reached by a binary prefix $z \in \{0, 1\}^{\leq r h_r}$ will be denoted $q_{r,S,z}$, with $q_{r,S,\varepsilon} = q_{r,S}$, and $\delta(q_{r,S,z}, a) = q_{r,S,za}$ ($|z| < r h_r$, $a \in \{0, 1\}$). Let

$$F_{r,S} = \{u_1 \cdots u_r : (u_1, \dots, u_r) \notin S\} \subseteq \{0, 1\}^{r h_r}.$$

Add one accepting state $T_{r,S}$, put $\delta(q_{r,S,z}, \#) = T_{r,S}$ ($z \in F_{r,S}$), and let $T_{r,S}$ loop on 0 and 1. Finally add $\delta(q_{r,S}, \lambda) = T_{r,S}$. Consequently every classifier root is live, and for every $z \in \{0, 1\}^{r h_r}$ and $v \in \{0, 1\}^*$,

$$\delta(q_{r,S}, z \# v) \in F \iff z \in F_{r,S}. \quad (10)$$

The additional λ -branch does not affect (10).

For each r , also create a length-gate root $B_r = B_{r,0}$ and states $B_{r,0}, B_{r,1}, \dots, B_{r,h_r}$, with $\delta(B_{r,j}, a) = B_{r,j+1}$ ($0 \leq j < h_r$, $a \in \{0,1\}$), where B_{r,h_r} is accepting and has no outgoing 0, 1-transition. Thus

$$\delta(B_r, u) \in F \iff u \in \{0,1\}^{h_r}. \quad (11)$$

It remains to make all of these roots accessible from one start state. Let

$$\mathcal{R}_n = \{q_{r,S} : 1 \leq r \leq k, S \subseteq (\{0,1\}^{h_r})^r\} \cup \{B_r : 1 \leq r \leq k\}$$

and put $J_n = |\mathcal{R}_n|$. Choose $d_n = \lceil \log_2 J_n \rceil$ and inject $\mathcal{R}_n$ into the fixed-length code set $\{\alpha, \beta\}^{d_n}$. Build the complete α/β -selector trie of depth d_n from a new start state q_0^* , and identify the selected leaves with the corresponding roots in $\mathcal{R}_n$. Unused leaves are left as dead states. Because all selector codewords have the same length, no selected root is an internal selector state. Every state in the resulting partial DFA $A_n^* = (Q_n^*, \Sigma^*, \delta_n^*, q_0^*, F_n^*)$ is accessible.

Lemma 4.21 (Size of the master DFA). *For fixed k ,*

$$|Q_n^*| = O_k(n2^n).$$

Proof. For fixed r , there are 2^{N_r} classifier roots. A complete binary trie of depth $r h_r$ has $2^{r h_r + 1} - 1 = 2N_r - 1$ states, and each classifier uses one additional accepting state. Hence the classifier bank has at most

$$\sum_{r=1}^k 2^{N_r} (2N_r + 1) \leq (2n + 1)k2^n = O_k(n2^n)$$

states by (9). Also $J_n = \sum_{r=1}^k 2^{N_r} + k = O_k(2^n)$. The complete selector trie has fewer than $2^{d_n + 1} \leq 4J_n$ states. Finally the k length gates use only $\sum_{r=1}^k (h_r + 1) = O_k(\log n)$ states. Combining the three bounds proves the claim. $\square$

Let $\Gamma_n^* = \Gamma_{A_n^*}$ be the FICS obtained from the automaton simulation of Section 4. Since A_n^* is accessible, Proposition 4.9 gives FCP and, because the alphabet Σ^* is fixed,

$$|\Gamma_n^*| = O_k(n2^n). \quad (12)$$

Rank-two candidate scaffolds. For $1 \leq r < k$, define a rank-two head pattern

$$G_{k,r}[x_1, \dots, x_k]$$

using neutral vertices

$$z_0, \dots, z_r, z_r^+, z_{r+1}, \dots, z_k.$$

For $1 \leq i \leq r$, let x_i have ports (z_{i-1}, z_i) ; let x_{r+1} have ports (z_r^+, z_{r+1}) ; and for $i > r + 1$, let x_i have ports (z_{i-1}, z_i) . Add the single ordinary edge $\{z_r, z_r^+\}$ labeled $\#$, and take the ordered interface to be $\iota(G_{k,r}) = (z_0, z_k)$. For $r = k$, use neutral vertices

$$z_0, \dots, z_k, z_k^+,$$

let x_i have ports (z_{i-1}, z_i) , add the ordinary $\#$ -edge $\{z_k, z_k^+\}$, and take interface $\iota(G_{k,k}) = (z_0, z_k^+)$. Then, for $u_i \in \{0,1\}^*$,

$$\text{Real}(G_{k,r}; x_i := P[u_i]) \cong_\iota \begin{cases} P[u_1 \cdots u_r \# u_{r+1} \cdots u_k], & r < k, \\ P[u_1 \cdots u_k \#], & r = k. \end{cases} \quad (13)$$

For every $S \subseteq (\{0,1\}^{h_r})^r$, define

$$\gamma_{r,S} : Q_{q_{r,S}}(G_{k,r}[x_1, \dots, x_k]) \leftarrow Q_{B_r}(S_{x_1}), \dots, Q_{B_r}(S_{x_k}). \quad (14)$$

Lemma 4.22 (Syntax and liveness of the bank candidates). *For fixed k , all candidates $\gamma_{r,S}$ are live admissible fixed-interface clauses in one bounded regime independent of n, r, S . They have exactly k distinct rank-two variable labels.*

Proof. Each $G_{k,r}$ contains exactly one variable-hyperedge occurrence labeled x_i for each $1 \leq i \leq k$, and the body of (14) contains exactly one star S_{x_i} with the same label and rank. Hence the occurrence-to-body-star correspondence is the bijection $x_i \longleftrightarrow Q_{B_r}(S_{x_i})$, $1 \leq i \leq k$, so the exact fixed-interface condition is satisfied.

The head pattern has rank two, matching the rank-two predicate $Q_{q_{r,S}}$, and every body predicate Q_{B_r} also has rank two. Thus one may take $s = t = k$ and $w = 2$. For fixed k , choose a constant d_k^* and let the finite family $\mathcal{H}_{d_k^*}$ contain $\{G_{k,1}, \dots, G_{k,k}\}$ together with the finitely many automaton frames E_a , $P[a]$, and W for the fixed alphabet Σ^* . Hence the bounded parameters and head-frame family are independent of n, r, S .

Finally, Q_{B_r} is live by (11). Also $\delta(q_{r,S}, \lambda) = T_{r,S} \in F_n^*$, so Lemma 4.4 gives $\Gamma_n^* \models Q_{q_{r,S}}(P[\lambda])$. Thus every predicate occurring in $\gamma_{r,S}$ is live. $\square$

Let $\mathfrak{B}_k^*$ be the fixed presentation-level subregime of the bounded FICS class of [14] obtained by fixing the finite alphabets of the master-automaton construction, taking $\Delta = 2$, $s = t = k$, $w = 2$, choosing $d = d_k^*$, and fixing the finite admissible head-frame family $\mathcal{H}_{d_k^*}$ described above. All these choices depend only on k , not on n, r, S , or the eventual clause budget m . Then every Γ_n^* and every candidate $\gamma_{r,S}$ lie in this same subregime.

For $S \subseteq (\{0, 1\}^{h_r})^r$, write

$$S^\# = \{(P[u_1], \dots, P[u_r]) : (u_1, \dots, u_r) \in S\} \subseteq A_r^r.$$

Lemma 4.23 (Exact fixed-target bank relation). *For every $1 \leq r \leq k$ and every $S \subseteq (\{0, 1\}^{h_r})^r$,*

$$\boxed{V_{\Gamma_n^*, \gamma_{r,S}} = S^\# \times A_r^{k-r}.$$

For $r = k$, the second factor is understood as a singleton empty product.

Proof. Let $\bar{K} = (K_1, \dots, K_k) \in \mathcal{G}_2^k$. If some K_i is not interface-isomorphic to a binary path $P[u_i]$ of length h_r , then Lemma 4.4 together with (11) gives $\Gamma_n^* \not\models Q_{B_r}(K_i)$. Hence not all body atoms are refutable and $V_{\Gamma_n^*, \gamma_{r,S}}(\bar{K}) = 0$.

Now suppose $K_i \cong_{\iota} P[u_i]$, $u_i \in \{0, 1\}^{h_r}$ ($1 \leq i \leq k$). All body atoms are then refutable. By (13), the realized head is the rank-two path whose word is

$$w = \begin{cases} u_1 \cdots u_r \# u_{r+1} \cdots u_k, & r < k, \\ u_1 \cdots u_k \#, & r = k. \end{cases}$$

The word before the separator has length rh_r . By (10) and Lemma 4.4,

$$\Gamma_n^* \models Q_{q_{r,S}}(P[w]) \iff u_1 \cdots u_r \in F_{r,S} \iff (u_1, \dots, u_r) \notin S.$$

The candidate-rejection convention is body-positive/head-negative. Therefore the tuple is rejected exactly when $(u_1, \dots, u_r) \in S$, proving the displayed equality. The private λ -transition cannot affect this calculation because the candidate-head word begins with 0 or 1. $\square$

Lemma 4.24 (Context compatibility of the fixed-target boxes). *For each classifier root $q_{r,S}$, let $v_{r,S} \in \{\alpha, \beta\}^{d_n}$ be its selector access word, and let v_r be the selector access word of B_r . Use the anchored FCP contexts $C_{v_{r,S}}$ and C_{v_r} . Then*

$$A_r^k \subseteq \Omega_{\Gamma_n^*, \gamma_{r,S}, C}.$$

Proof. Take $K_i = P[u_i] \in A_r$. Every body star realizes to $P[u_i]$. The anchored context C_{v_r} has matching rank-two interface labels and no ordinary edge joining its two interface vertices, so $C_{v_r} \odot P[u_i]$ is defined. The realized head in (13) is also a rank-two path, and the same argument shows that $C_{v_r, S} \odot \text{Real}(G_{k, r}; x_i := P[u_i])$ is defined. Thus all body and head compositions required in the definition of $\Omega_{\Gamma_n^*, \gamma_{r, S}, C}$ are defined. $\square$

Theorem 4.25 (Simultaneously sharp fixed-target profiles). *For every fixed $k \geq 2$, the bounded regime $\mathfrak{B}_k^*$ above has, for all sufficiently large clause budgets m , a target*

$$\Gamma_m^* \in \mathfrak{B}_k^*, \quad |\Gamma_m^*| \leq m,$$

such that simultaneously for every $1 \leq r \leq k$,

$$d_r(\Gamma_m^*) = \Theta_{k, r}((\log(m+1))^{1/r}).$$

The lower bounds already hold using live admissible candidates. Moreover,

$$\log_2 |\mathcal{V}_k(\Gamma_m^*)| = \Theta_k(\log(m+1)).$$

Consequently, for every $1 \leq r \leq k$,

$$(d_r(\Gamma_m^*))^r \asymp_{k, r} \log_2 |\mathcal{V}_k(\Gamma_m^*)| \asymp_k \log(m+1).$$

Proof. First work with the target Γ_n^* . Fix r . If $r < k$, choose any $a_r \in A_r^{k-r}$ and fix the last $k-r$ coordinates to a_r ; for $r = k$, no coordinates are fixed. By Lemma 4.23, as $S \subseteq A_r^r$ varies, the corresponding fiber restrictions of $V_{\Gamma_n^*, \gamma_{r, S}}$ realize every subset of A_r^r . Hence $d_r(\Gamma_n^*) \geq D_r$. Using (9), $d_r(\Gamma_n^*)^r \geq D_r^r = N_r > \frac{n}{2^r}$. Thus, simultaneously for all r ,

$$d_r(\Gamma_n^*) = \Omega_{k, r}(n^{1/r}). \quad (15)$$

By Lemma 4.21 and Proposition 4.9, there is a constant C_k such that $|\Gamma_n^*| \leq C_k n 2^n$ for all sufficiently large n . Define $B_n = \lceil C_k n 2^n \rceil$. Then $\log_2 B_n = n + \log_2 n + O_k(1) = \Theta_k(n)$. For an arbitrary sufficiently large budget m , choose the largest n such that $B_n \leq m$, and set $\Gamma_m^* = \Gamma_n^*$. Since $m < B_{n+1} = O_k((n+1)2^{n+1})$, we still have $n = \Theta_k(\log(m+1))$. Combining this with (15) gives $d_r(\Gamma_m^*) = \Omega_{k, r}((\log(m+1))^{1/r})$. The matching upper bound is Proposition 3.6, applied to the fixed regime $\mathfrak{B}_k^*$. This proves the profile statement.

For the entropy lower bound, use $r = 1$. Then $N_1 = D_1 \geq n/2$. The relations $V_{\Gamma_n^*, \gamma_{1, S}} = S^\# \times A_1^{k-1}$, $S \subseteq A_1$, are pairwise distinct, so $|\mathcal{V}_k(\Gamma_n^*)| \geq 2^{N_1} \geq 2^{n/2}$. Hence $\log_2 |\mathcal{V}_k(\Gamma_m^*)| = \Omega_k(\log(m+1))$. The fixed-target entropy upper bound in Proposition 3.6 gives the matching $O_k(\log(m+1))$ estimate. The final displayed equivalence follows. $\square$

Corollary 4.26 (Learner-side realization inside one target). *Fix the master target Γ_n^* . There is a sample threshold N_n such that, for every learner stage $j \geq N_n$, the current residual set contains, simultaneously for every $1 \leq r \leq k$ and every $P[u] \in A_r$, a representation $\sigma_{r, u}$ satisfying $\text{can}(\sigma_{r, u}) \cong_l P[u]$. Moreover, at every update stage $j \geq N_n$ that occurs in the learner run, the predicate basis F contains representatives whose canonical graphs are the explicit anchored FCP contexts C_{v_r} and $C_{v_r, S}$ for all bank predicates used at the fixed master size n . Consequently, at every such update stage, for every r and every $S \subseteq A_r^r$, the representative translation of $\gamma_{r, S}$ is an actual bounded candidate of Algorithm 3, its Cartesian-product loop tests every tuple in A_r^r , and the rejection bits on that box are exactly $S^\# \times A_r^{k-r}$. This is again a conditional realization statement: no existence claim is made for an update stage $j \geq N_n$.*

Proof. For every $u \in \{0, 1\}^{h_r}$, $\Gamma_n^* \models Q_{B_r}(P[u])$, so the anchored FCP context C_{v_r} gives the positive start-language graph $C_{v_r} \odot P[u]$. By Lemma 4.7, once these graphs have appeared in the positive sample, $\text{BRep}_2(D_j)$ contains residual representations whose canonical graphs are all members of A_r .

The same positive graphs expose the gate contexts C_{v_r} themselves by Lemma 4.8. For every classifier root $q_{r,S}$, the private liveness transition gives $\Gamma_n^* \models Q_{q_{r,S}}(P[\lambda])$, so $C_{v_{r,S}} \odot P[\lambda]$ is also a positive start-language graph; Lemma 4.8 extracts the explicit context $C_{v_{r,S}}$ from it. For fixed n , there are only finitely many indices r, S, u , hence a complete positive presentation contains all these positive anchored graphs from some common stage N_n onward.

Algorithm 4 sets $R = \text{BRep}_2(D_j)$ at every stage. At an update stage it also resets $F = \text{BRep}_2(D_j)$. Thus every update stage $j \geq N_n$ that occurs in the learner run simultaneously contains all required residual and predicate representations. The entire box lies in the context-compatible domain by Lemma 4.24. Theorem 2.6 then identifies the non-fact Cartesian-product admission test of Algorithm 3 with the target-relative rejection bit on A_r^k , and Lemma 4.23 gives the exact relation $S^\sharp \times A_r^{k-r}$. $\square$

5 Discussion and Open Problems

The two sharp theorems identify distinct information levels of the same target-induced candidate-rejection mechanism. Globally, both the target presentation and the target-relative candidate may vary, and the available description space has logarithm $\Theta(m \log m)$ in the rich regime. Conditional on one fixed target, only target-relative candidate choice remains, reducing the exact information scale to $\Theta(\log m)$. The distinction is therefore not merely between two proof techniques but between two different sources of information in the target-induced candidate class:

$$\boxed{(d_r^{\mathfrak{B}_k}(m))^r \asymp m \log(m+1)} \quad \text{versus} \quad \boxed{(d_r(\Gamma_m^*))^r \asymp \log(m+1)}.$$

Both scales are simultaneously sharp for all $1 \leq r \leq k$.

5.1 Graph-language, presentation, and candidate-rejection complexity

Previous FGS VC/PAC results analyze generated graph languages $L(\Gamma, p) \subseteq \mathcal{G}$. Our object lies at the level of candidate evaluation: $V_{\Gamma, \gamma} \subseteq \mathcal{G}_2^k$ records simultaneous substitutions on which a target-relative candidate is rejected. Refutability makes this more than a purely syntactic relation, but it remains presentation-relative: equivalent presentations of the same start language need not induce the same candidate class. Actual target clauses do not contribute nontrivial relations, by Lemma 2.3.

5.2 Relation to the FICS candidate test

The bridge in Theorem 2.6 is deliberately a representative translation rather than a set inclusion. Target predicates and learner representative predicates are different syntactic alphabets. At a stage where the required representatives belong to the current predicate basis F , the bounded-candidate construction of Algorithm 3 in [14] includes the translated fixed-interface clause in its candidate set, while its non-fact Cartesian-product admission test evaluates the residual-family rejection bit studied here. For live target predicates, Lemmas 4.15 and 4.17 of [14] give the corresponding representative-exposure mechanism at sufficiently large characteristic update stages. Conversely, the learner may contain boundary representatives that do not correspond to target predicates, so we do not identify the full sample-induced candidate space with $\mathcal{V}_k^{\mathfrak{B}}(m)$.

Three levels of candidate space. It is useful to keep three levels distinct. First, $\mathcal{V}_k^{\mathfrak{B}}(m)$ is a global target-induced class in which both the target presentation and an admissible candidate over its active predicates may vary. Second, $\mathcal{V}_k(\Gamma)$ fixes the target and varies only such target-induced candidates. Third, at a concrete learner stage, Algorithm 3 constructs candidates over the sample-dependent basis $F \cup \{\rho_\emptyset\}$. This third space may be strictly larger because F can contain boundary representatives unrelated to target predicates.

The present capacity theorems concern the first two levels. The direct realization results establish an embedding of the relevant live target-induced candidates into the third level at stages where their representatives have been exposed; they do not identify the three classes. Determining the capacity of the full sample-induced learner-state candidate space is a separate problem.

The extremal construction sharpens this connection. For each prescribed labeling S , Lemma 4.7 gives an explicit ordered-boundary specification that recovers every path block $P[u]$ from the positive anchored graph $C_b \odot P[u]$. Hence the current residual set eventually contains all block substitutions in the extremal box. Theorem 4.15 then supplies, at stages with the required predicate representative, an actual Algorithm 3 candidate whose rejection bits on that box are exactly S . Lemma 4.12 verifies that the whole box lies inside the domain on which the FCP observation bridge is valid. Shattering is obtained across the family of targets; it would be incorrect to say that one fixed target or one learner run shatters the box. This stagewise realization is conditional: representative exposure guarantees the required contents of R and, at a sufficiently late update stage, of F , but it does not by itself guarantee that such a post-threshold update stage occurs.

5.3 FCP terminology and tuple-based distributional learning

Our FCP is the FICS finite context property of [14]. The strong/weak/very weak finite-context and finite-kernel hierarchy for string grammars [8] is conceptually related through the distributional-learning tradition, but no formal identification is made here. Likewise, tuple-based MCFG work studies multidimensional nonterminals and substitutability [3, 16]. The present arity $|X(\gamma)|$ instead counts simultaneous independent substitution slots in one candidate clause. A systematic transfer between these two notions of arity would be interesting, but is outside the present theorem package.

5.4 Global versus fixed-target fiber profiles

The global sharp theorem varies both the target Γ and the candidate γ . For a single fixed target, Proposition 3.6 gives $d_r(\Gamma) = O((\log(m+1))^{1/r})$. This logarithmic scale is operationally closer to the family of candidate tests available during one run on one target.

Theorem 4.25 closes the sharp-order question: there is one bounded target family Γ_m^* for which $d_r(\Gamma_m^*)^r = \Theta_{k,r}(\log(m+1))$ simultaneously for every $1 \leq r \leq k$. Thus the global and fixed-target scales are both sharp, but at different information levels:

$$(m \log m)^{1/r} \quad \text{versus} \quad (\log m)^{1/r}.$$

The construction also shows that the fixed-target lower bounds can be witnessed by live candidates and, at any update stage occurring after the relevant representative-exposure threshold, by actual candidate tests inside one learner run on the same target. This is a conditional stagewise realization statement, not an assertion that a post-threshold update stage must occur.

What remains open is not simultaneous sharpness but the finer structure of all possible fixed-target profile vectors.

Problem 5.1 (Fixed-target profile-vector characterization). Characterize, up to asymptotic equivalence or sharp inequalities, the vectors $(d_1(\Gamma), \dots, d_k(\Gamma))$ that can arise from one bounded FCP-FICS target Γ of clause budget m . In particular, what tradeoffs, if any, are forced between different fiber dimensions beyond the coordinatewise bounds $d_r(\Gamma)^r = O(\log(m+1))$?

5.5 Growing arity

All asymptotics above fix k .

Problem 5.2 (Joint m, k dependence). Determine explicit bounds for $d_r(m, k)$ when the number of independent clause variables is allowed to grow with the grammar.

5.6 Description length instead of clause count

The clause-budget formulation assumes a fixed bounded-head regime. If arbitrary head graphs are allowed, one clause can encode unbounded syntax.

Problem 5.3 (Description-length complexity). Let $\|\Gamma\|$ charge for the entire encoded syntax of a FICS. Determine the higher-arity VC profile as a function of $\|\Gamma\|$, without fixing a finite head-frame family in advance.

An information-theoretic upper bound linear in the bit description length is natural, but the sharp lower-order factors depend on the chosen encoding.

5.7 Minimal interfaces

The lower bound already uses only paths, rank-two substitution interfaces, fixed finite alphabets, and maximum graph degree two.

Problem 5.4 (Minimal interface rank). Can the $m \log(m+1)$ entropy lower bound be realized with rank-one interfaces, or is rank two necessary for transmitting ordered finite-state information through substitution?

5.8 Algorithmic consequences

The present results are capacity bounds, not runtime or membership-query lower bounds. The learner searches only the residual tuples exposed by the current positive sample, whose exposure rate is not controlled by the present VC analysis.

Problem 5.5 (Clause-test compression). Can the fiber profile be exploited to reduce the product search over $R^{X(\gamma)}$ used to validate candidate clauses, for example by constructing small witness sets, adaptive tests, or parameterized procedures for systems with small realized fiber profiles?

This would connect the combinatorial analysis directly back to the product search in the FICS learning algorithm [14].

5.9 A possible general template

The proof also suggests a broader mechanism that is not pursued as a theorem here. Suppose a bounded clause formalism has at most $2^{O(m \log(m+1))}$ canonical descriptions at syntax budget m , and can realize an n -state finite automaton with $O(n)$ syntactic overhead while preserving the relevant observation semantics. The same counting-versus-finite-state-realization argument suggests an r -th-root box-capacity scale of order $(m \log(m+1))^{1/r}$. Determining natural hypotheses under which this becomes a formal transfer theorem is left for future work.

5.10 Conclusion

The candidate test of the FICS learner carries a higher-arity geometry that is not visible from the generated start language alone. Its distinct variable labels index an explicit Cartesian substitution search, and the pointwise body-positive/head-negative outcomes of that search form the candidate-rejection relations studied here.

The main structural conclusion is a separation between two sharp information regimes. Conditional on one target presentation of clause budget m , variation within the target-induced candidate class has only logarithmic entropy, and this bound is simultaneously sharp in every fiber dimension: $(d_r(\Gamma_m^*))^r \asymp_{k,r} \log_2 |\mathcal{V}_k(\Gamma_m^*)| \asymp_k \log(m+1)$. One master target attains these bounds for all $1 \leq r \leq k$ at once by storing finite banks of live classifier predicates.

If the target presentation is also allowed to vary, the available information increases to the global scale $(d_r^{\mathfrak{B}_k}(m))^r \asymp \log_2 |\mathcal{V}_k^{\mathfrak{B}_k}(m)| \asymp m \log(m+1)$. Thus the global $m \log m$ term reflects target-presentation information in addition to candidate choice; it should not be interpreted as the amount of variation available inside one fixed learner target.

These two capacity levels concern target-induced candidate classes. They should not be confused with the potentially larger sample-dependent candidate space over the learner basis F . Our Algorithm 3 realization results show that the relevant live target-induced candidates occur inside that learner space once their representatives have been exposed, but no capacity characterization of the full learner-state space is claimed here.

Attaining both sharp scales inside the FICS framework requires an explicit realization architecture: fixed-length residual compression, linear-size partial-DFA simulation, preservation of FCP, live predicates, exact length gates, anchored contexts, ordered-boundary residual exposure, and translation into the candidate syntax actually tested by Algorithm 3. At any update stage occurring after the required finite exposure threshold, the translated learner candidate realizes the prescribed pointwise rejection test. This learner-side statement is conditional on the occurrence of such an update stage.

The resulting picture connects finite-state description complexity to graph grammar learning not merely through an abstract counting analogy, but through an exact transfer into the learner-compatible candidate geometry of bounded fixed-interface clause systems. Conceptually, it also gives a concrete grammatical-inference realization of the product-space perspective motivating higher-dimensional PAC_n -learning: the higher arity comes directly from the learner’s simultaneous substitution slots, while the claims established here remain deliberately at the level of sharp shattering capacity rather than a general PAC_n characterization.

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